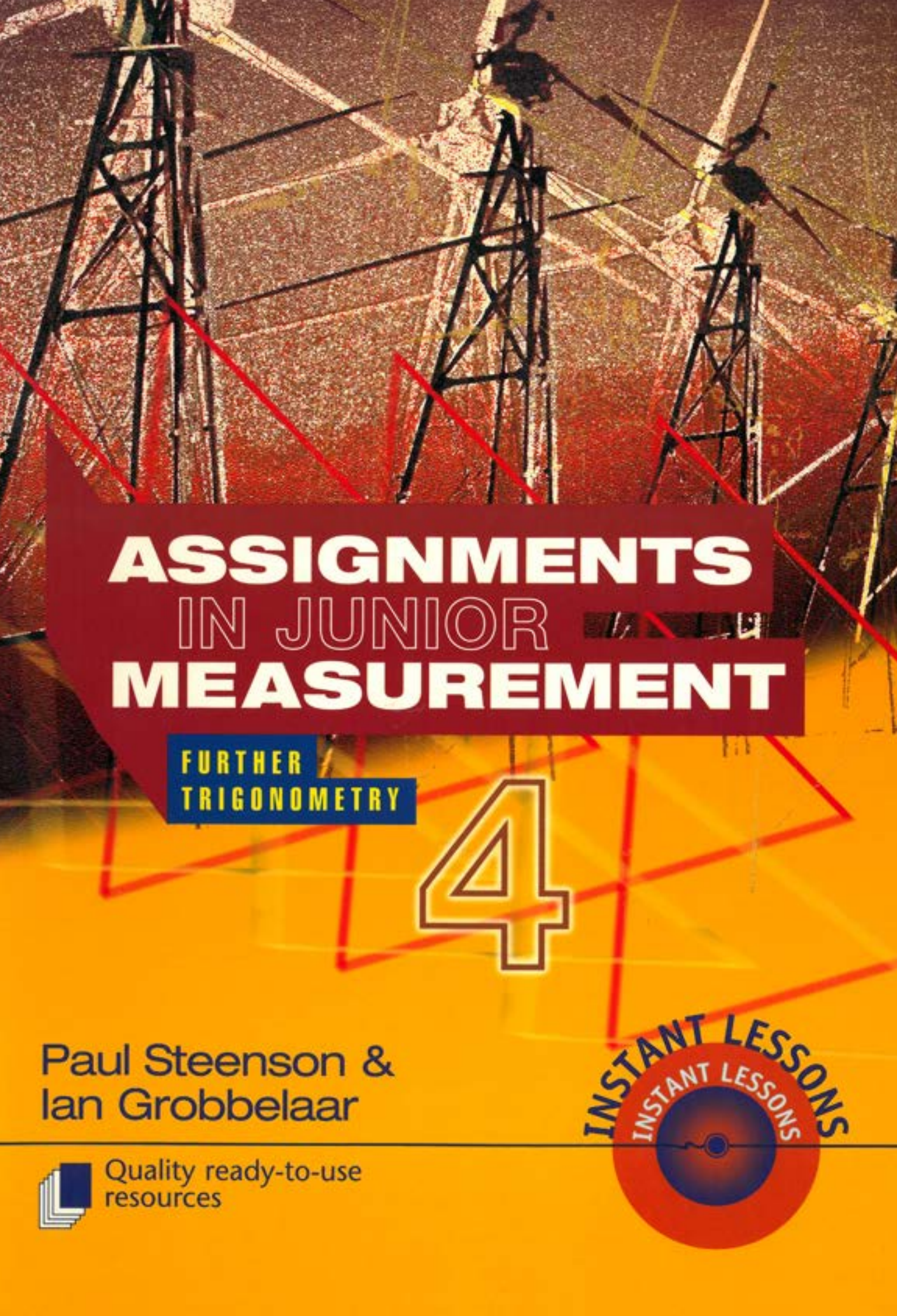




ASSIGNMENTS IN JUNIOR MEASUREMENT

FURTHER
TRIGONOMETRY

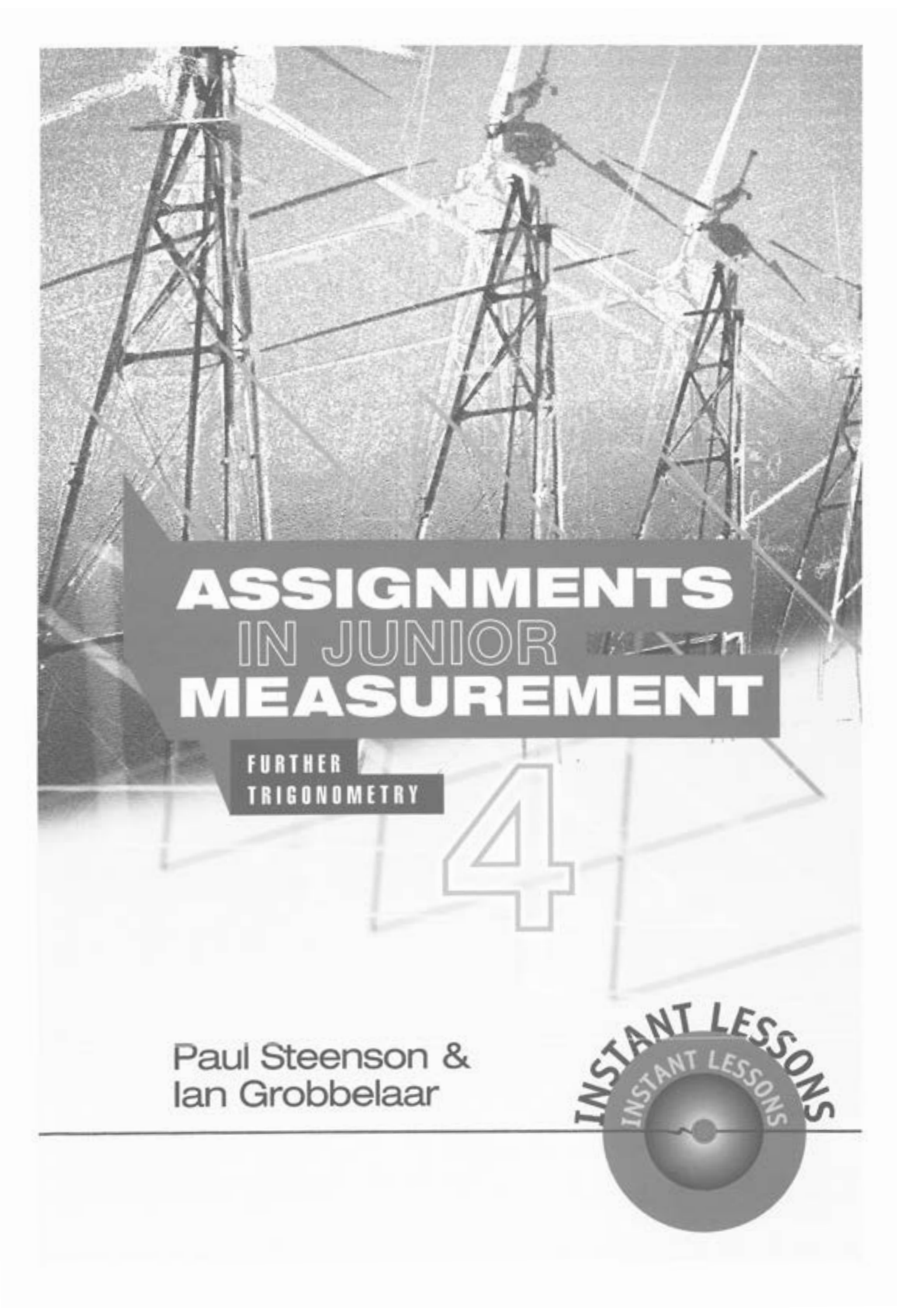
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Paul Steenson &
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ASSIGNMENTS IN JUNIOR **MEASUREMENT**

**FURTHER
TRIGONOMETRY**

4

Paul Steenson &
Ian Grobbelaar



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Project editor: Susan Watson

Editor: Ruth Schultz

Design, formatting and diagrams: Ruth Schultz

Cover design: Gail McManus Graphics

Printed in Singapore by Green Giant Press

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Introduction

This book provides a systematic and comprehensive coverage of selected topics from the junior mathematics curriculum. It is independent of any textbook and is not linked to any syllabus.

Each worksheet follows a format comprising three sections:

Section A consists of a set of introductory questions or review of previous knowledge which can be used as a lesson starter or 'warm up' exercise.

Section B provides skills development or drill with graded questions in the focus area of the worksheet.

Section C provides further practice in the form of different types of questions, a riddle, challenge or extension questions.

Review worksheets are also included in each topic area.

There are minimal examples or explanations as teachers have different methods or working that they would like their students to follow.

Most worksheets could be used by students in a range of years or courses. The teacher may decide to use only one or two sections from a worksheet in a lesson or as a homework assignment. The worksheets are designed to be flexible to meet the needs of the teacher and students.

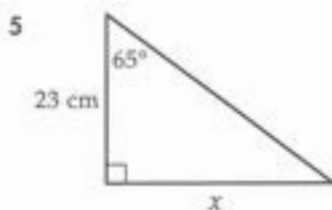
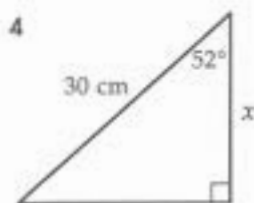
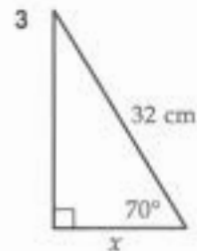
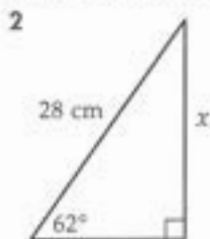
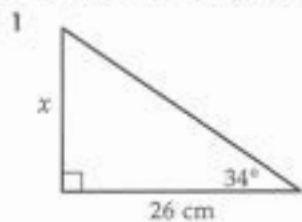
A time required for students to complete each worksheet is not suggested as this will vary significantly depending on the year level of the students and their level of mathematical skill. The teacher is the best person to make this judgement.

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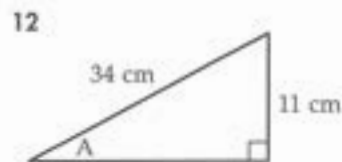
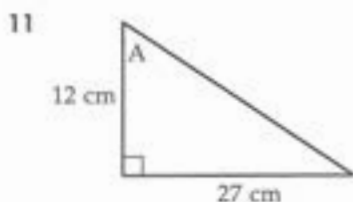
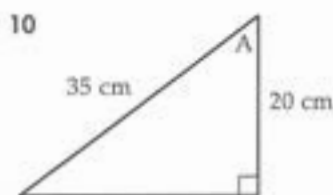
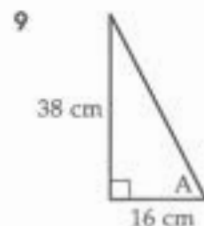
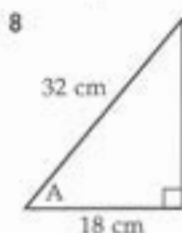
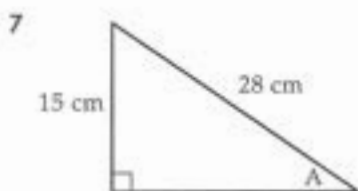
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Review: trigonometry of right-angled triangles 1.1

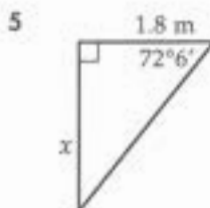
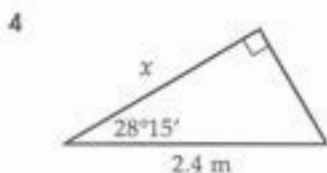
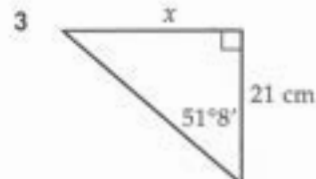
A Calculate the length of side x correct to 1 decimal place.



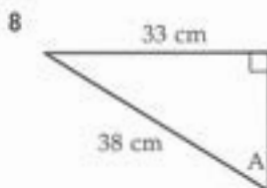
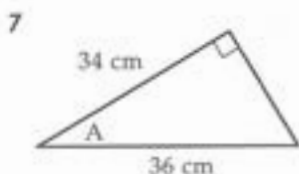
Calculate the size of $\angle A$ to the nearest degree.



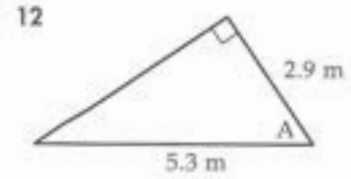
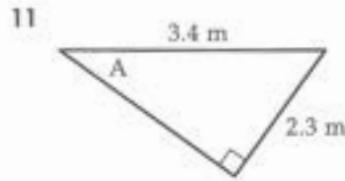
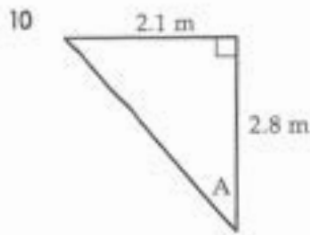
B Calculate the length of side x correct to 1 decimal place.



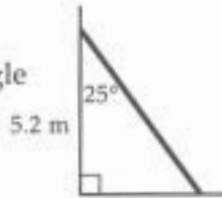
Find the size of $\angle A$ in degrees and minutes.



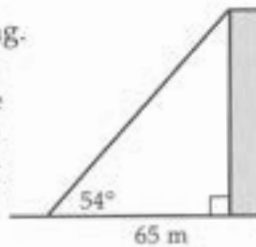
Find the size of $\angle A$ in degrees and minutes.



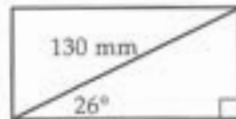
- C** 1 A ladder reaches 5.2 m up a vertical wall, making an angle of 25° with the wall. How far is the foot of the ladder from the base of the wall?



- 2 Loi is standing 65 metres from the base of a building. He measures the angle of elevation of the top of the building as 54° . Calculate the height of the building correct to 1 decimal place.



- 3 The diagonal of a rectangle is 130 mm long. It makes an angle of 26° with the longer sides. Calculate the length of the rectangle to the nearest millimetre.



- 4 A ramp rises 2.85 metres and makes an angle of 18° with the horizontal ground. How long is the ramp?



- 5 A road rises 80 metres in a straight length of 950 metres. Find the angle of slope to the nearest degree.

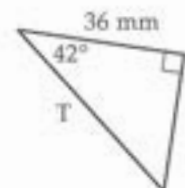
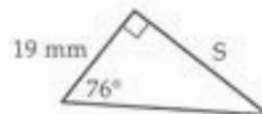
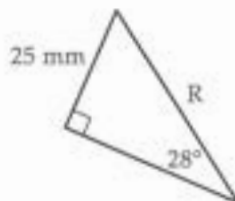
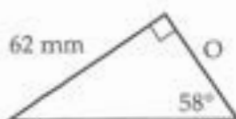
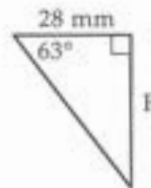
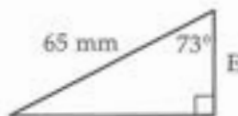
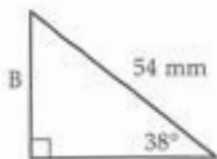
- 6 When the sun's rays are at an angle of 43° to the horizontal, a tower casts a shadow 35.4 metres long. Calculate the height of the tower correct to 1 decimal place.

- 7 A 4.8-metre ladder leans against a vertical wall. The foot of the ladder is 1.5 m from the wall. What angle (in degrees and minutes) does the ladder make with the ground?

- 8 A boat is 350 m from the base of a vertical cliff. From the top of the cliff, the angle of depression to the boat is $17^\circ 45'$. Calculate the height of the cliff to the nearest metre.

Riddle: What do you get when you cross a snowman with a vampire?

Calculate the length of each side marked with a pronumeral to the nearest millimetre. Then write the pronumeral above each correct answer.



55	53	39	76	48	33	35	48	19	

All answers are in millimetres.

Exact ratios for 30° , 45° and 60°

2

A Use your calculator to evaluate each ratio correct to 3 decimal places.

1 $\tan 45^\circ$

2 $\cos 60^\circ$

3 $\sin 30^\circ$

4 $\sin 45^\circ$

5 $\cos 65^\circ$

6 $\tan 60^\circ$

7 $\cos 30^\circ$

8 $\sin 60^\circ$

Use your calculator to find the size of $\angle A$ to the nearest degree.

9 $\sin A = 0.707$

10 $\tan A = 1$

11 $\cos A = 0.707$

12 $\sin A = 0.866$

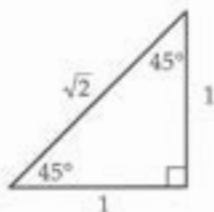
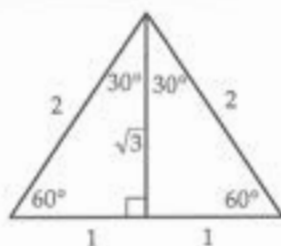
13 $\cos A = 0.5$

14 $\sin A = 0.5$

15 $\cos A = 0.866$

16 $\sin A = \frac{1}{\sqrt{2}}$

B



EXAMPLE: Find the exact ratio for $\sin 45^\circ$.
From these diagrams we can see that:

$$\sin 45^\circ = \frac{\text{opp}}{\text{hyp}} = \frac{1}{\sqrt{2}}$$

Use the diagrams above (not a calculator) to evaluate each ratio.

1 $\sin 60^\circ$

2 $\cos 60^\circ$

3 $\tan 60^\circ$

4 $\tan 45^\circ$

5 $\cos 30^\circ$

6 $\tan 30^\circ$

7 $\sin 30^\circ$

8 $\sin 45^\circ$

C Use the diagrams above (not a calculator) to find the size of angle A.

1 $\cos A = \frac{\sqrt{3}}{2}$

2 $\sin A = \frac{\sqrt{3}}{2}$

3 $\tan A = 1$

4 $\cos A = \frac{1}{\sqrt{2}}$

5 $\tan A = \sqrt{3}$

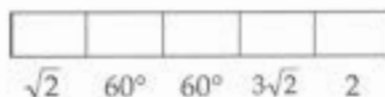
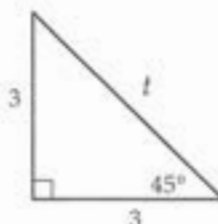
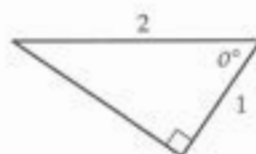
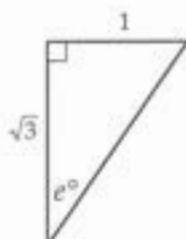
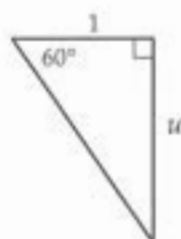
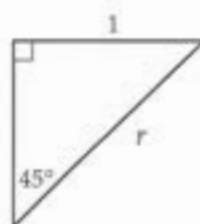
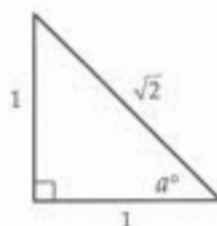
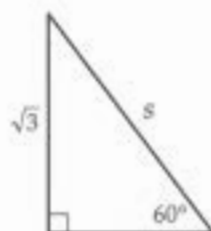
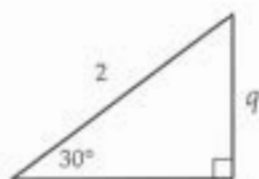
6 $\sin A = \frac{1}{2}$

7 $\tan A = \frac{1}{\sqrt{3}}$

8 $\cos A = \frac{1}{2}$

Riddle: What was different about the mathematics tree?

Find the value of each pronumeral. Then write the pronumerals above each correct answer.



A Use your calculator to evaluate each ratio correct to 3 decimal places.

- | | | | |
|--------------------|--------------------|--------------------|--------------------|
| 1 $\tan 25^\circ$ | 2 $\sin 37^\circ$ | 3 $\cos 15^\circ$ | 4 $\tan 42^\circ$ |
| 5 $\sin 74^\circ$ | 6 $\tan 65^\circ$ | 7 $\cos 28^\circ$ | 8 $\sin 56^\circ$ |
| 9 $\tan 70^\circ$ | 10 $\cos 68^\circ$ | 11 $\tan 34^\circ$ | 12 $\sin 44^\circ$ |
| 13 $\cos 53^\circ$ | 14 $\sin 16^\circ$ | 15 $\tan 83^\circ$ | 16 $\cos 37^\circ$ |

B When you use the trigonometric ratios, the angle does not need to be an acute angle (between 0° and 90°). We can find the sine, cosine or tangent ratio for angles of any magnitude, positive or negative.

Use your calculator to evaluate each ratio of an obtuse angle correct to 3 decimal places.

- | | | | |
|-------------------------|-------------------------|-------------------------|-------------------------|
| 1 $\sin 125^\circ$ | 2 $\cos 105^\circ$ | 3 $\tan 120^\circ$ | 4 $\cos 155^\circ$ |
| 5 $\sin 160^\circ$ | 6 $\tan 130^\circ$ | 7 $\cos 100^\circ$ | 8 $\sin 95^\circ$ |
| 9 $\cos 132^\circ$ | 10 $\sin 115^\circ$ | 11 $\tan 105^\circ$ | 12 $\sin 152^\circ$ |
| 13 $\sin 140^\circ$ | 14 $\cos 126^\circ$ | 15 $\tan 161^\circ$ | 16 $\cos 100^\circ$ |
| 17 $\tan 156^\circ$ | 18 $\sin 95^\circ$ | 19 $\cos 146^\circ$ | 20 $\sin 107^\circ$ |
| 21 $\sin 137^\circ$ | 22 $\cos 162^\circ$ | 23 $\tan 121^\circ$ | 24 $\cos 93^\circ$ |
| 25 $\sin 134^\circ 52'$ | 26 $\tan 103^\circ 15'$ | 27 $\cos 145^\circ 27'$ | 28 $\sin 150^\circ 45'$ |
| 29 $\cos 103^\circ 53'$ | 30 $\sin 98^\circ 25'$ | 31 $\cos 105^\circ 50'$ | 32 $\tan 161^\circ 36'$ |
| 33 $\sin 127^\circ 35'$ | 34 $\cos 126^\circ 5'$ | 35 $\tan 152^\circ 55'$ | 36 $\sin 102^\circ 20'$ |
| 37 $\tan 106^\circ 12'$ | 38 $\sin 115^\circ 26'$ | 39 $\cos 173^\circ 39'$ | 40 $\tan 104^\circ 48'$ |

C Use your calculator to evaluate each ratio correct to 4 decimal places.

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| 1 $\sin 200^\circ$ | 2 $\cos 195^\circ$ | 3 $\tan 260^\circ$ | 4 $\cos 305^\circ$ |
| 5 $\sin 225^\circ$ | 6 $\tan 312^\circ$ | 7 $\cos 210^\circ$ | 8 $\sin 350^\circ$ |
| 9 $\cos 290^\circ$ | 10 $\sin 325^\circ$ | 11 $\tan 230^\circ$ | 12 $\sin 210^\circ$ |
| 13 $\sin 285^\circ$ | 14 $\tan 256^\circ$ | 15 $\cos 345^\circ$ | 16 $\tan 205^\circ$ |
| 17 $\cos 282^\circ$ | 18 $\sin 197^\circ$ | 19 $\sin 336^\circ$ | 20 $\cos 321^\circ$ |
| 21 $\tan 349^\circ$ | 22 $\cos 236^\circ$ | 23 $\sin 258^\circ$ | 24 $\tan 304^\circ$ |
| 25 $\cos (-25^\circ)$ | 26 $\sin (-37^\circ)$ | 27 $\tan (-42^\circ)$ | 28 $\cos (-74^\circ)$ |
| 29 $\tan (-63^\circ)$ | 30 $\sin (-70^\circ)$ | 31 $\cos (-56^\circ)$ | 32 $\sin (-82^\circ)$ |

Find the size of the obtuse angle A (between 90° and 180°) to the nearest degree.

- | | | | |
|----------------------|----------------------|----------------------|----------------------|
| 33 $\sin A = 0.56$ | 34 $\sin A = 0.72$ | 35 $\sin A = 0.32$ | 36 $\sin A = 0.45$ |
| 37 $\cos A = -0.25$ | 38 $\cos A = -0.64$ | 39 $\cos A = -0.38$ | 40 $\cos A = -0.72$ |
| 41 $\tan A = -0.65$ | 42 $\tan A = -0.85$ | 43 $\tan A = -1.25$ | 44 $\tan A = -1.63$ |
| 45 $\sin A = 0.92$ | 46 $\cos A = -0.42$ | 47 $\tan A = -0.56$ | 48 $\sin A = 0.84$ |
| 49 $\cos A = -0.378$ | 50 $\sin A = 0.613$ | 51 $\cos A = -0.186$ | 52 $\tan A = -2.576$ |
| 53 $\tan A = -0.752$ | 54 $\sin A = 0.951$ | 55 $\tan A = -1.856$ | 56 $\cos A = -0.446$ |
| 57 $\cos A = -0.905$ | 58 $\tan A = -0.475$ | 59 $\sin A = 0.399$ | 60 $\sin A = 0.986$ |
| 61 $\cos A = -0.577$ | 62 $\tan A = -3.473$ | 63 $\sin A = 0.215$ | 64 $\cos A = -0.813$ |

The sine rule: finding the length of a side 4.1

A Use your calculator to evaluate each expression correct to 2 decimal places.

1 $\sin 56^\circ$

2 $\sin 78^\circ$

3 $\sin 48^\circ 26'$

4 $\sin 63^\circ 48'$

5 $\sin 105^\circ$

6 $\sin 129^\circ$

7 $\sin 116^\circ 35'$

8 $\sin 140^\circ 15'$

9 $14 \sin 28^\circ$

10 $3.2 \sin 110^\circ$

11 $28 \sin 93^\circ 55'$

12 $5.4 \sin 137^\circ 33'$

13 $\frac{48}{\sin 35^\circ}$

14 $\frac{4.6}{\sin 102^\circ}$

15 $\frac{37}{\sin 115^\circ 42'}$

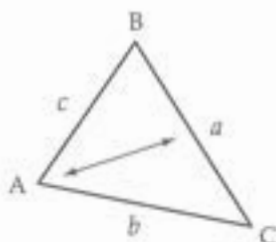
16 $\frac{2.9}{\sin 53^\circ 9'}$

B

The **sine rule** states:

In any triangle ABC,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

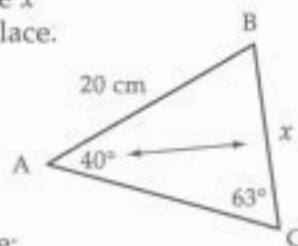


That is, the lengths of the sides are in the same proportion as the sine ratios of the angles opposite the sides.

Note: it is normal practice to name the vertices and matching angles with capital letters and the opposite sides in the corresponding lower case letters.

EXAMPLE:

Find the length of side x correct to 1 decimal place.

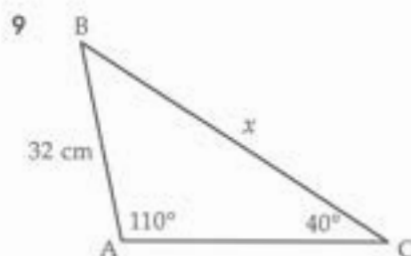
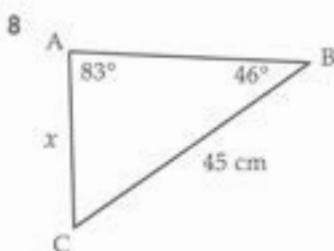
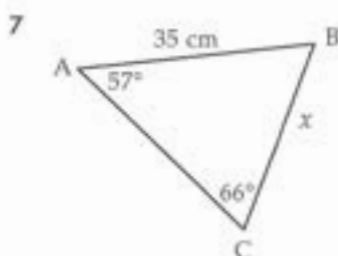
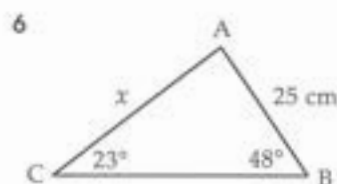
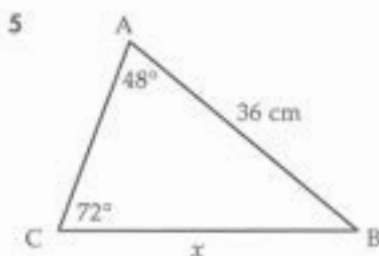
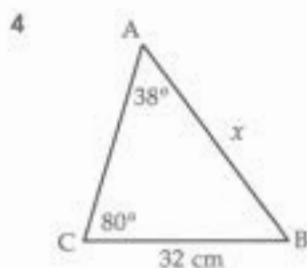
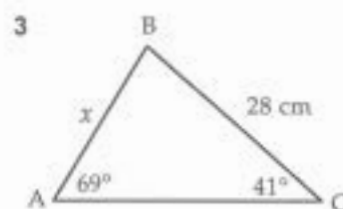
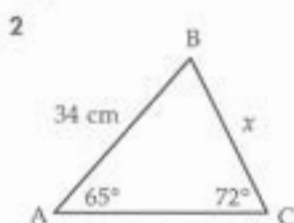
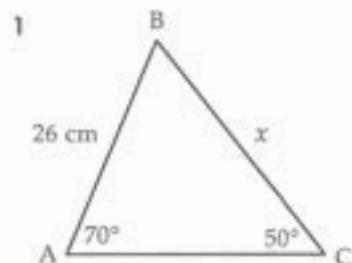


Applying the sine rule:

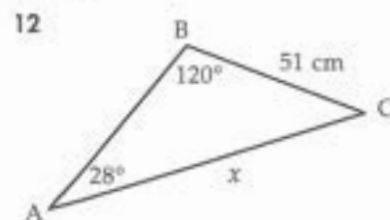
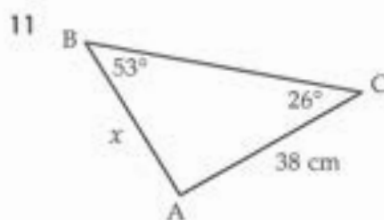
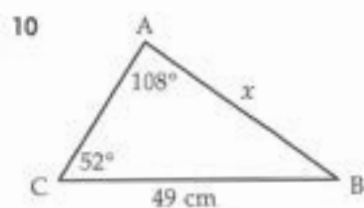
$$\frac{x}{\sin 40^\circ} = \frac{20}{\sin 63^\circ}$$

rearrange: $x = \frac{20 \times \sin 40^\circ}{\sin 63^\circ}$
 $= 14.4 \text{ cm}$

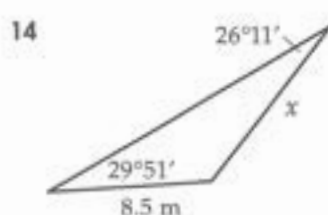
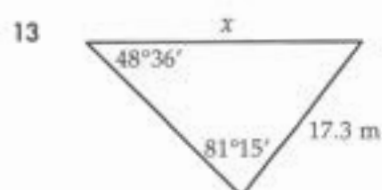
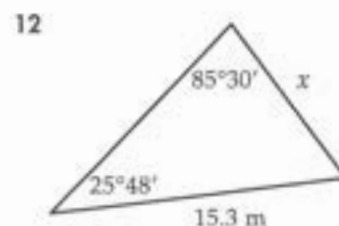
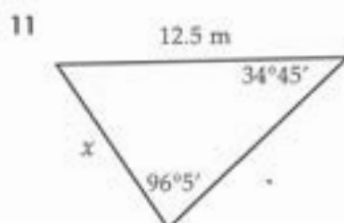
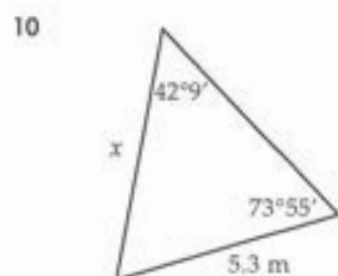
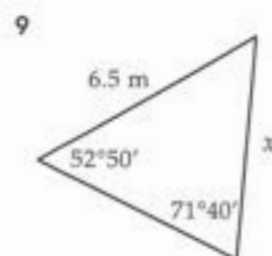
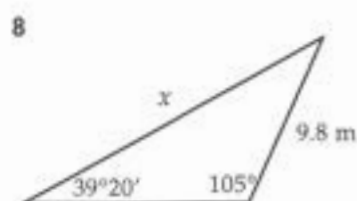
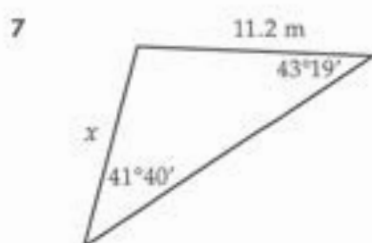
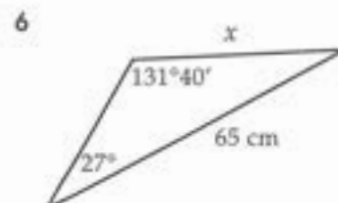
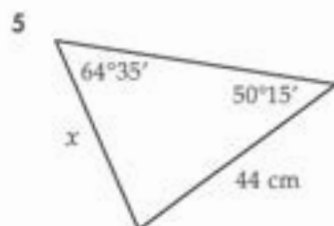
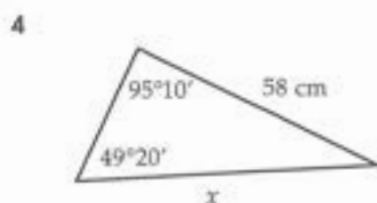
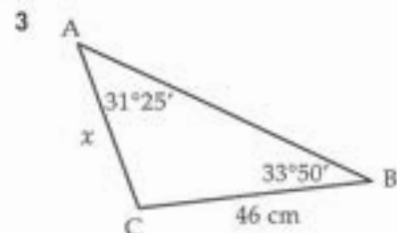
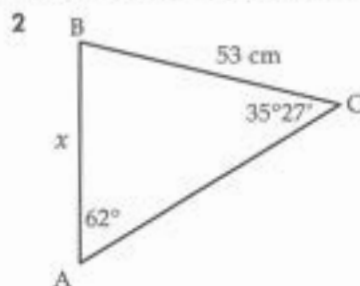
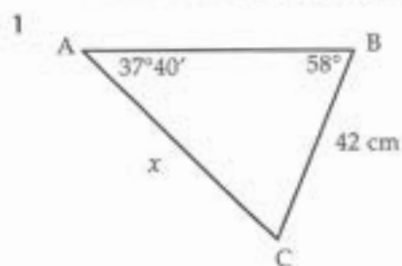
Use the sine rule to calculate the length of side x correct to 1 decimal place.



Use the sine rule to calculate the length of side x correct to 1 decimal place.



C Use the sine rule to calculate the length of side x correct to 1 decimal place.



The sine rule: finding the size of an angle 5.1

A Use your calculator to find the size of the **acute** angle A to the nearest degree.

- | | | | |
|--------------------|--------------------|--------------------|--------------------|
| 1 $\sin A = 0.914$ | 2 $\sin A = 0.438$ | 3 $\sin A = 0.755$ | 4 $\sin A = 0.225$ |
| 5 $\sin A = 0.573$ | 6 $\sin A = 0.192$ | 7 $\sin A = 0.696$ | 8 $\sin A = 0.971$ |

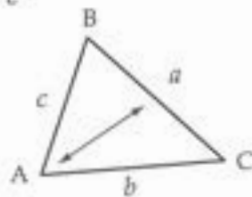
Use your calculator to find the size of the **obtuse** angle A to the nearest degree.

- | | | | |
|---------------------|---------------------|---------------------|---------------------|
| 9 $\sin A = 0.819$ | 10 $\sin A = 0.996$ | 11 $\sin A = 0.642$ | 12 $\sin A = 0.422$ |
| 13 $\sin A = 0.952$ | 14 $\sin A = 0.276$ | 15 $\sin A = 0.324$ | 16 $\sin A = 0.788$ |

B

To find the size of an **angle**, use this form of the **sine rule**.

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$



EXAMPLE: Find the size of angle A to the nearest degree.

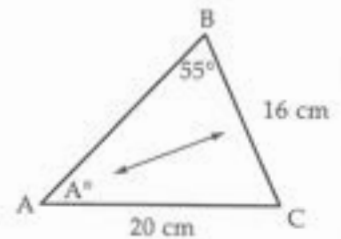
Applying the sine rule:

$$\frac{\sin A}{16} = \frac{\sin 55^\circ}{20}$$

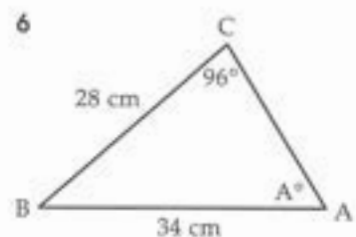
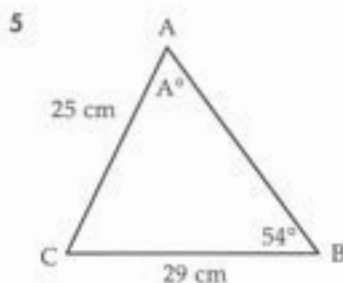
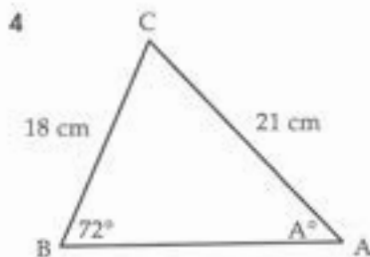
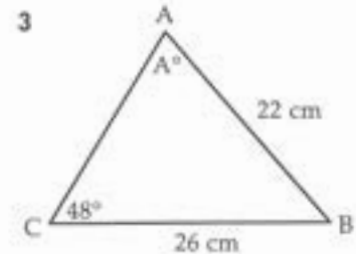
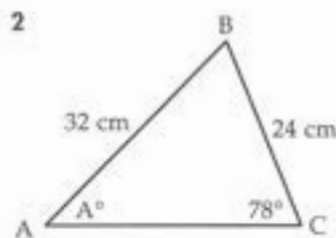
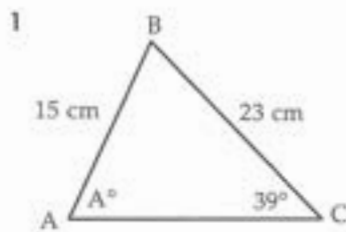
$$\text{rearrange: } \sin A = \frac{16 \times \sin 55^\circ}{20}$$

$$= 0.6553\dots$$

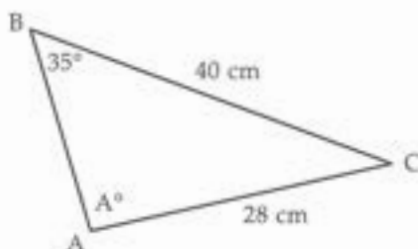
$$A = 41^\circ$$



Use the sine rule to find the size of the **acute** angle A to the nearest degree.



When you use the sine rule to find the size of an **obtuse** angle, you must subtract the acute angle from 180° .



EXAMPLE: Find the size of angle A to the nearest degree.

Applying the sine rule: $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$

$$\frac{\sin A}{40} = \frac{\sin 35^\circ}{28}$$

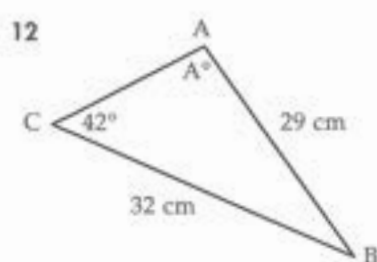
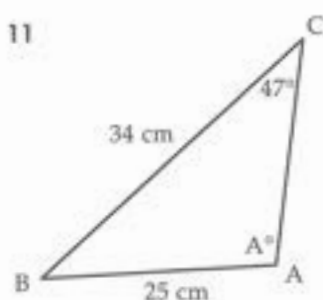
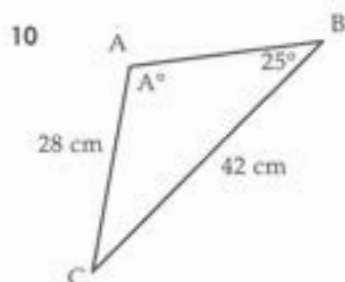
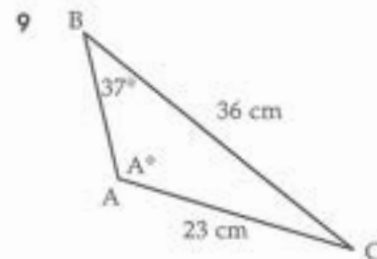
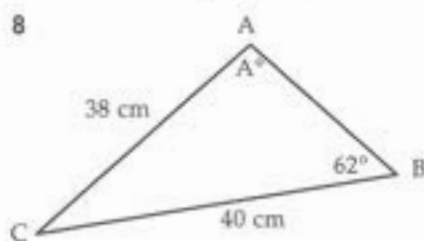
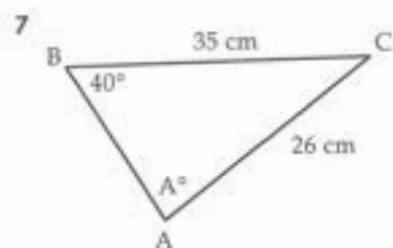
$$\text{rearrange: } \sin A = \frac{40 \times \sin 35^\circ}{28}$$

$$= 0.81939\dots$$

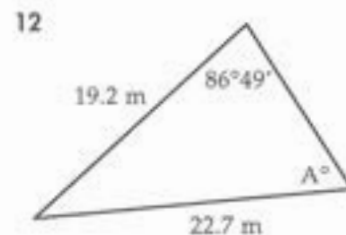
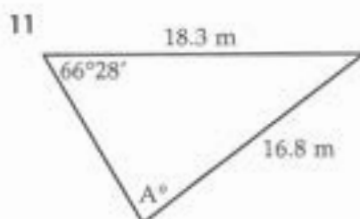
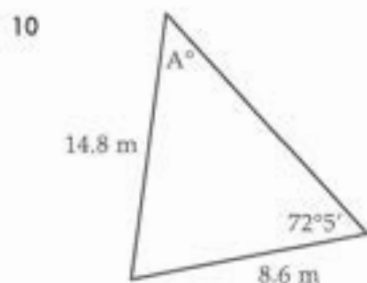
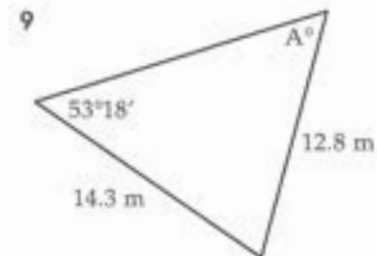
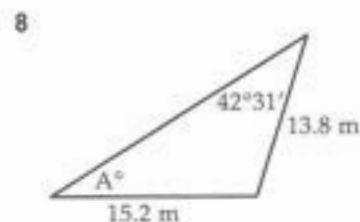
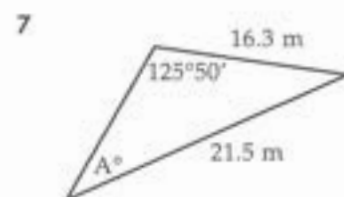
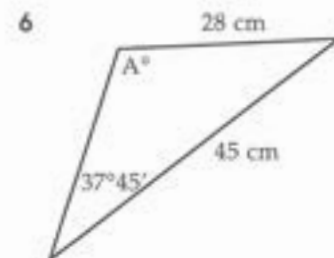
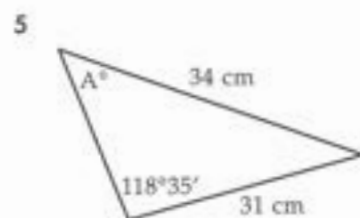
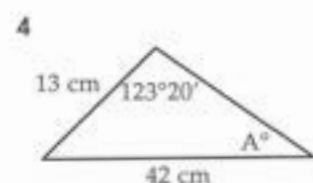
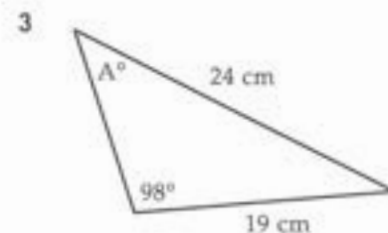
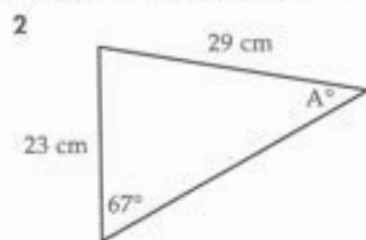
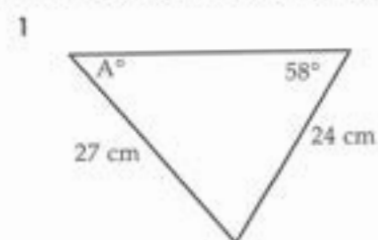
$$\text{calculator gives: } A = 55^\circ$$

$$\text{The obtuse angle is: } 180^\circ - 55^\circ = 125^\circ$$

Use the sine rule to find the size of the obtuse angle A to the nearest degree.



C Use the sine rule to find the size of angle A in degrees and minutes.



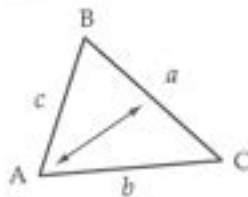
The sine rule: mixed questions

6.1

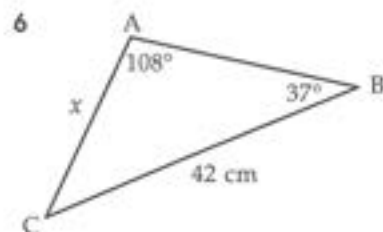
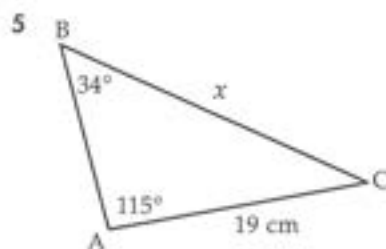
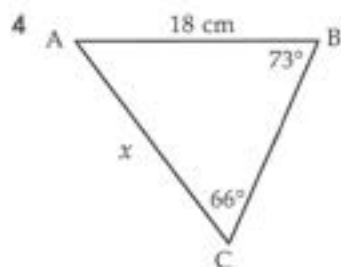
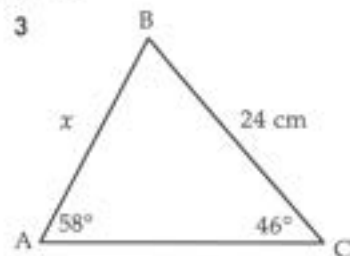
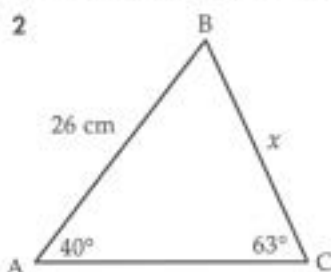
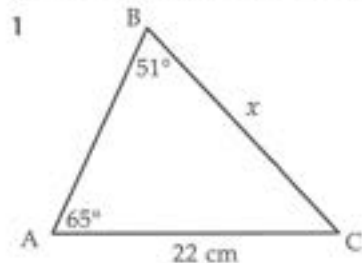
A

The sine rule states that in any triangle ABC:

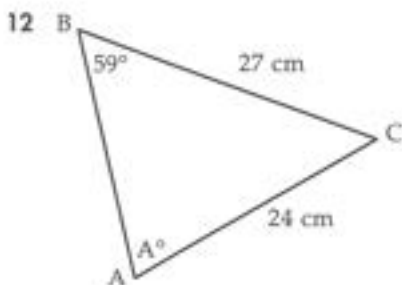
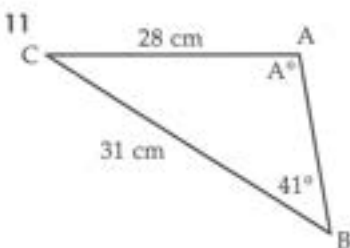
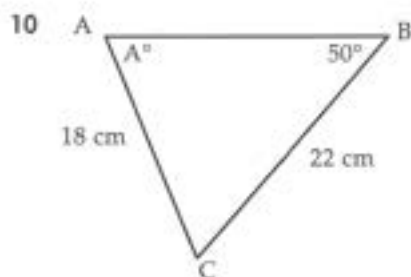
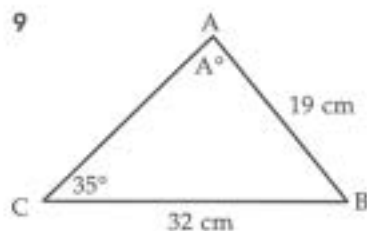
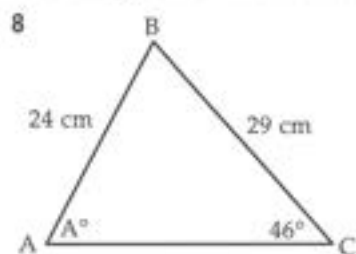
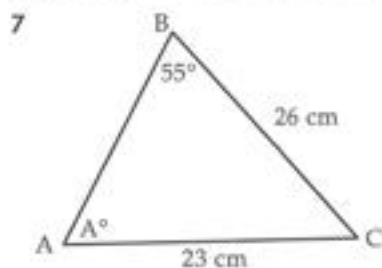
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$



Use the sine rule to calculate the length of side x correct to 1 decimal place.

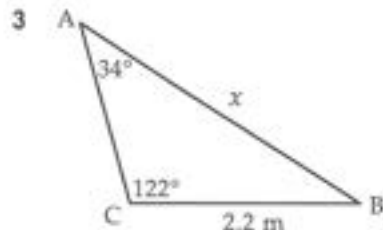
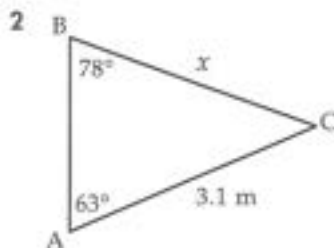
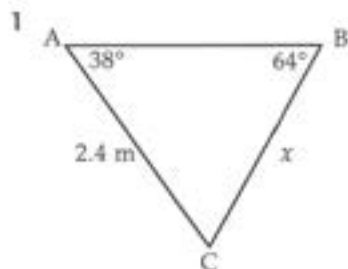


Use the sine rule to calculate the size of angle A to the nearest degree.

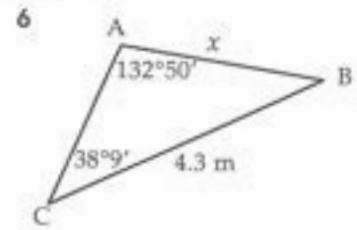
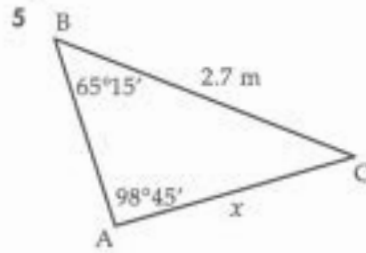
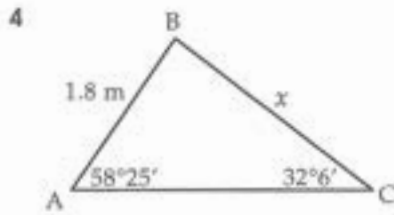


B

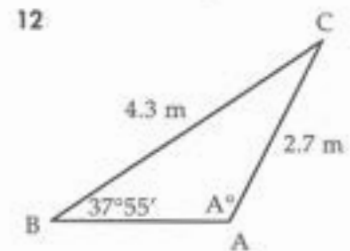
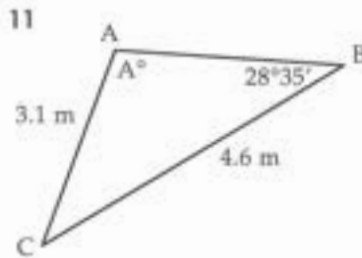
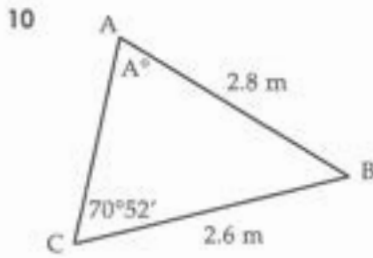
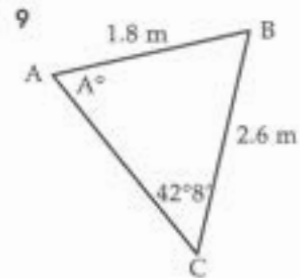
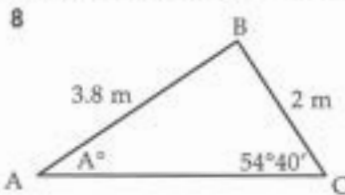
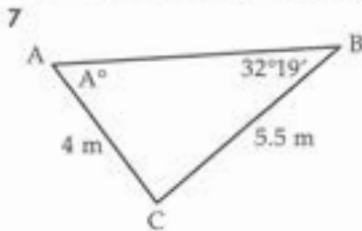
Use the sine rule to calculate the length of side x correct to 1 decimal place.



Use the sine rule to calculate the length of side x correct to 1 decimal place.



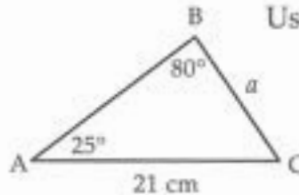
Use the sine rule to find the size of angle A in degrees and minutes.



C

EXAMPLE:

In the triangle ABC, $A = 25^\circ$,
 $B = 80^\circ$ and $b = 21$ cm.
 Find the length of side a
 to 1 decimal place.



Using the sine rule:

$$\frac{a}{\sin 25^\circ} = \frac{21}{\sin 80^\circ}$$

$$\text{rearrange: } a = \frac{21 \times \sin 25^\circ}{\sin 80^\circ}$$

$$= 9.0 \text{ cm}$$

For each question, draw a triangle ABC and label it with the given information. Then calculate the length of side a correct to 1 decimal place.

1 $A = 48^\circ$, $B = 72^\circ$ and $b = 15$ cm

2 $A = 62^\circ$, $B = 53^\circ$ and $b = 17$ cm

3 $A = 112^\circ$, $B = 31^\circ$ and $b = 13$ cm

4 $A = 121^\circ$, $C = 40^\circ$ and $c = 23$ cm

Find the size of angle A in degrees and minutes. (A is an acute angle.)

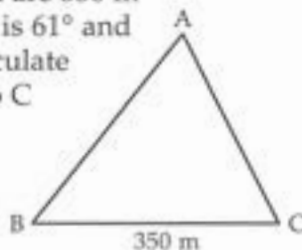
5 $B = 35^\circ$, $a = 15$ cm and $b = 13$ cm

6 $B = 61^\circ$, $a = 47$ cm and $b = 44$ cm

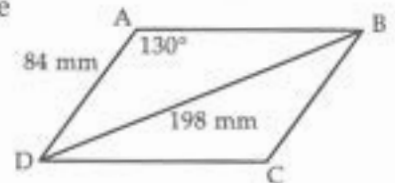
7 $B = 92^\circ$, $a = 32$ cm and $b = 48$ cm

8 $C = 42^\circ$, $a = 39$ cm and $c = 31$ cm

- 9 A steeple (A) is observed from the two points B and C, which are 350 m apart. The angle ABC is 61° and angle ACB is 73° . Calculate the distance from A to C to the nearest metre.



- 10 In the parallelogram ABCD, $AD = 84$ mm, angle DAB = 130° and the diagonal BD = 198 mm. Calculate the size of angle ABD to the nearest degree.



The cosine rule: finding the length of a side 7.1

A Use a calculator to evaluate each expression correct to 2 decimal places.

1 $\cos 82^\circ$

2 $\cos 27^\circ$

3 $\cos 16^\circ 32'$

4 $\cos 23^\circ 43'$

5 $\cos 108^\circ$

6 $\cos 149^\circ$

7 $\cos 122^\circ 13'$

8 $\cos 162^\circ 15'$

9 $23 \cos 54^\circ$

10 $5.4 \cos 139^\circ$

11 $29 \cos 15^\circ 56'$

12 $21.7 \cos 155^\circ 26'$

13 $\frac{67}{\cos 31^\circ}$

14 $\frac{5.6}{\cos 112^\circ}$

15 $\frac{53}{\cos 151^\circ 7'}$

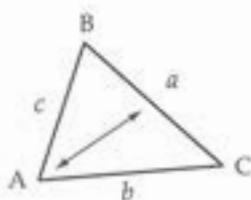
16 $\frac{13.7}{\cos 79^\circ 10'}$

B

The cosine rule states:

In any triangle ABC,

$$a^2 = b^2 + c^2 - 2bc \cos A$$



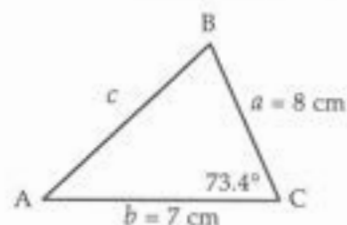
That is, use the cosine rule to find the length of an unknown side when you have the lengths of two of the sides and the size of the angle between those sides.

Also: $b^2 = a^2 + c^2 - 2ac \cos B$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

EXAMPLE:

Find AB.



Applying the cosine rule:

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$= 8^2 + 7^2 - 2(8)(7) \cos 73.4^\circ$$

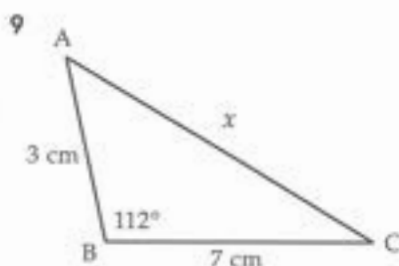
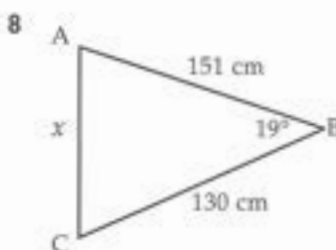
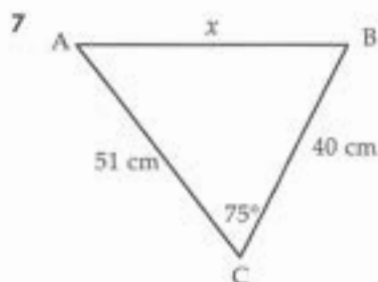
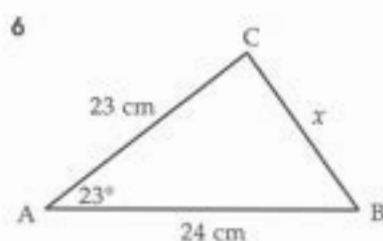
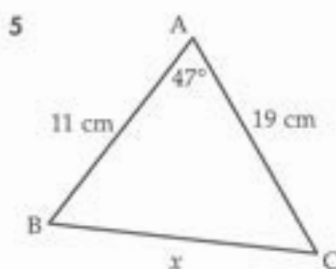
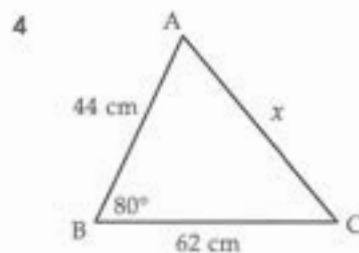
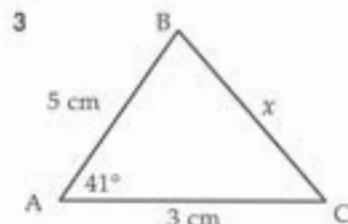
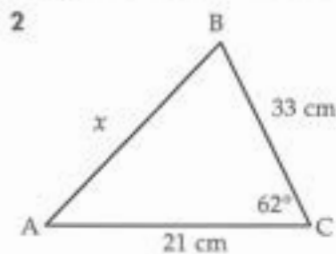
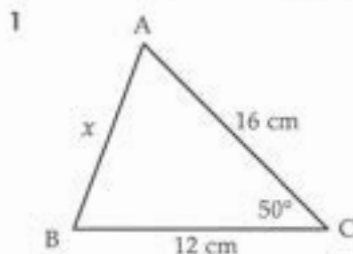
$$= 81.00290285 \dots$$

$$c = \sqrt{81.00290285}$$

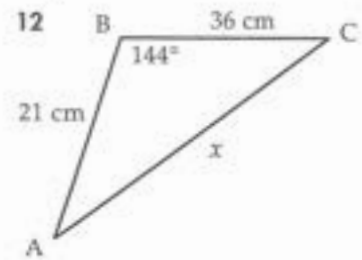
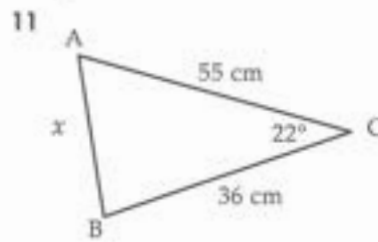
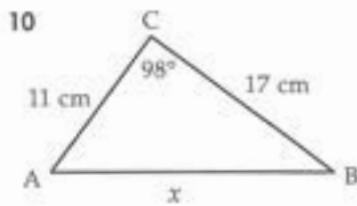
$$= 9$$

$$\therefore AB = 9 \text{ cm}$$

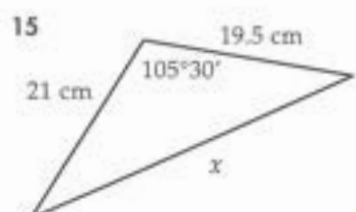
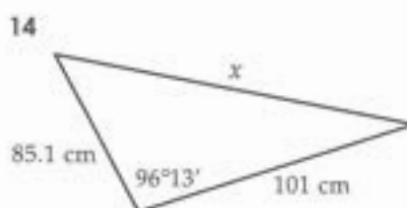
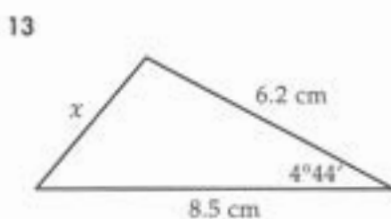
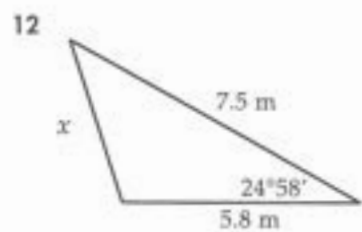
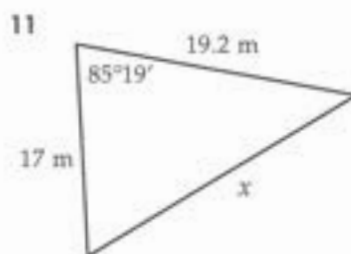
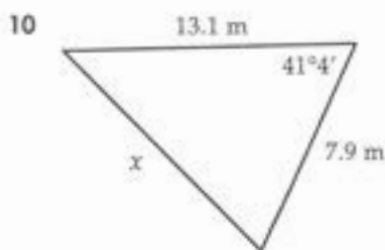
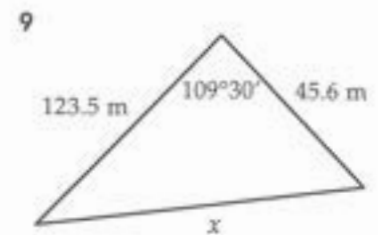
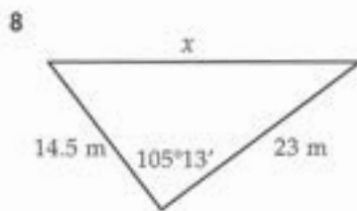
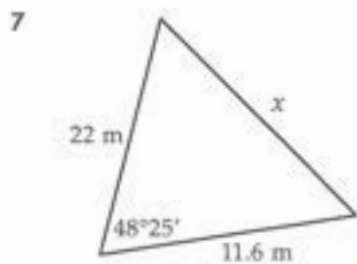
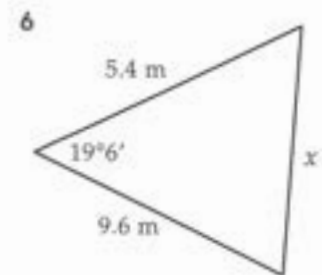
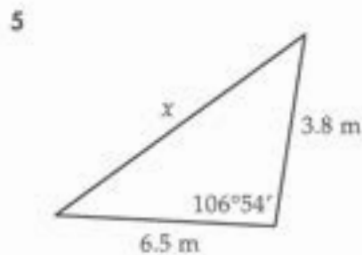
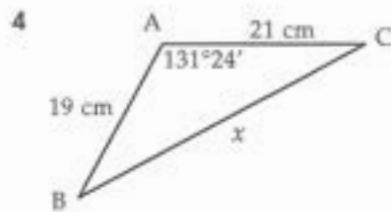
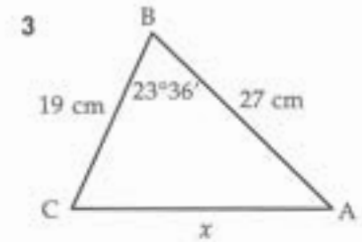
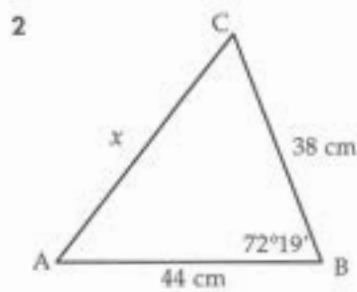
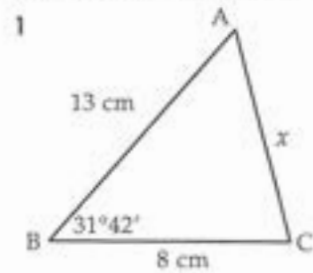
Use the cosine rule to calculate the length of side x correct to 1 decimal place.



Use the cosine rule to calculate the length of side x correct to 1 decimal place.



C Use the cosine rule to calculate the length of side x correct to 1 decimal place.



The cosine rule: finding the size of an angle 8.1

A Use your calculator to find the size of the **acute** angle A to the nearest degree.

1 $\cos A = 0.641$

2 $\cos A = 0.483$

3 $\cos A = 0.123$

4 $\cos A = 0.252$

5 $\cos A = 0.375$

6 $\cos A = 0.092$

7 $\cos A = 0.911$

8 $\cos A = 0.809$

Use your calculator to find the size of the **obtuse** angle A to the nearest degree.

9 $\cos A = -0.912$

10 $\cos A = -0.669$

11 $\cos A = -0.752$

12 $\cos A = -0.995$

13 $\cos A = -0.529$

14 $\cos A = -0.029$

15 $\cos A = -0.396$

16 $\cos A = -0.515$

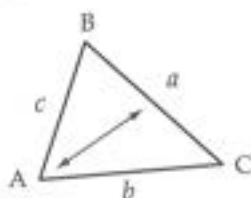
B

To find the size of an angle when you know the lengths of all 3 sides, use the cosine rule in this form:

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

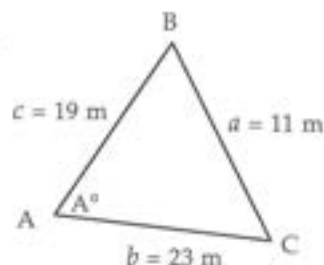
$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$



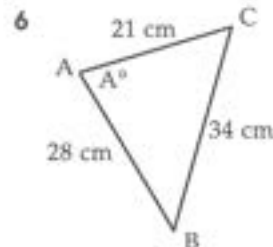
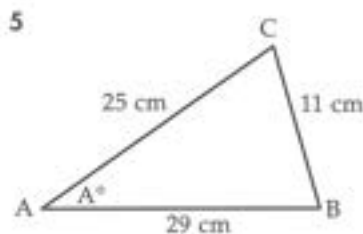
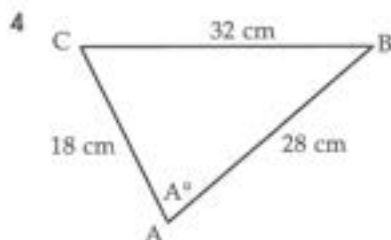
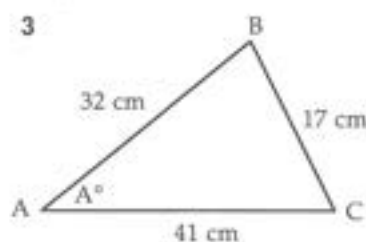
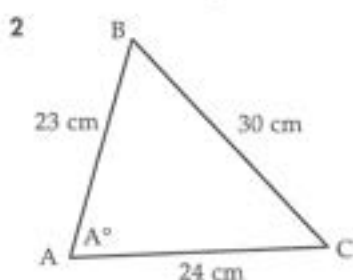
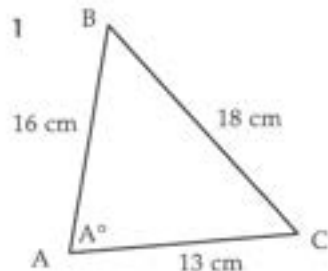
EXAMPLE:

Find the size of angle A to the nearest degree.

$$\begin{aligned}\cos A &= \frac{23^2 + 19^2 - 11^2}{2(23)(19)} \\ &= 0.8798627 \dots \\ \angle A &= 28^\circ\end{aligned}$$

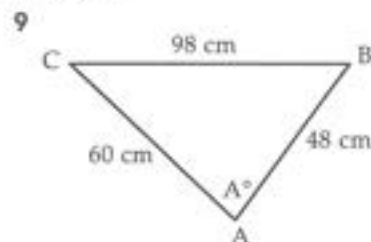
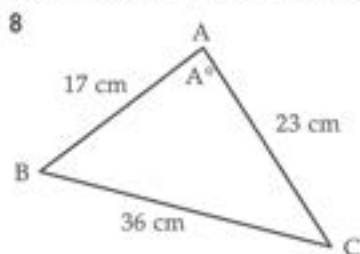
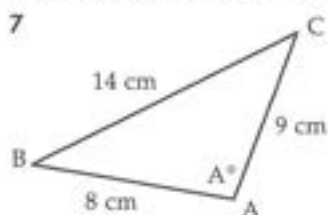


Use the cosine rule to find the size of the **acute** angle A to the nearest degree.

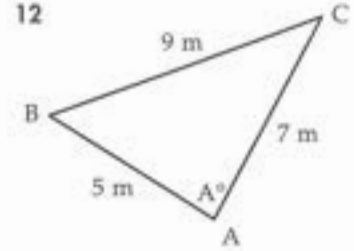
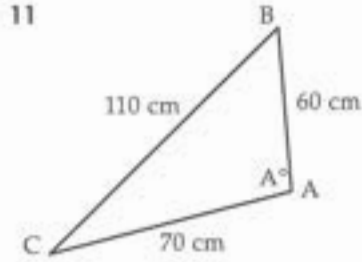
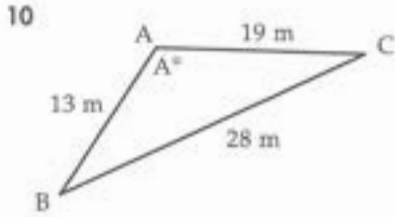


Remember that when using the cosine rule, if the ratio is **negative** then the angle is **obtuse**.

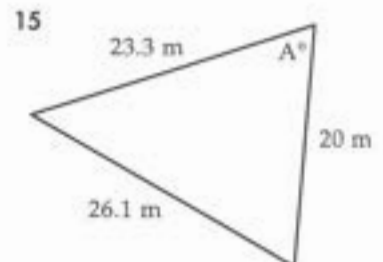
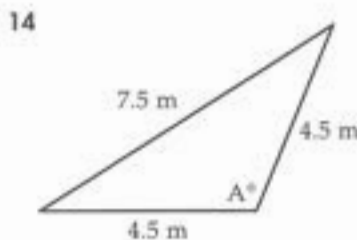
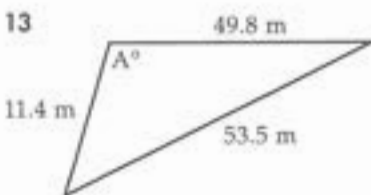
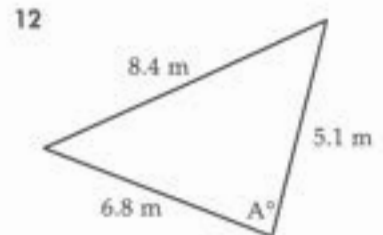
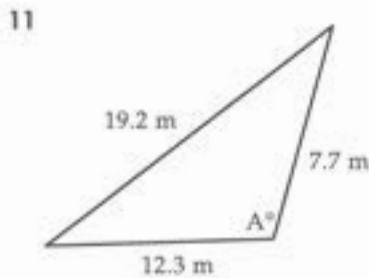
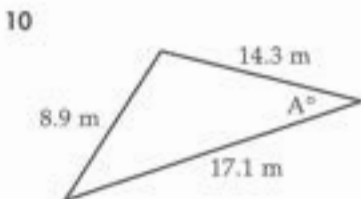
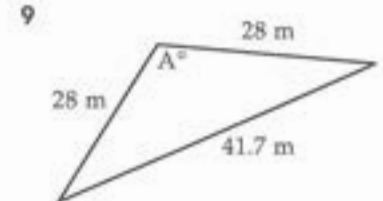
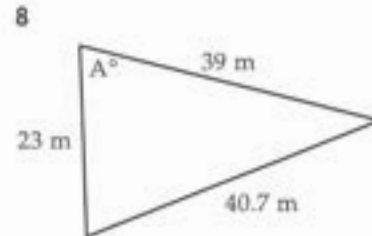
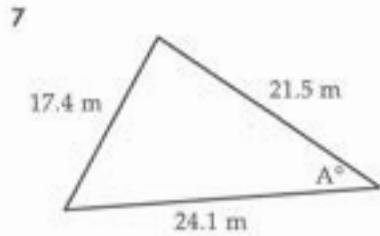
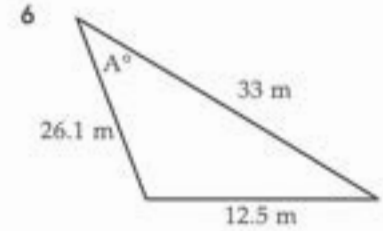
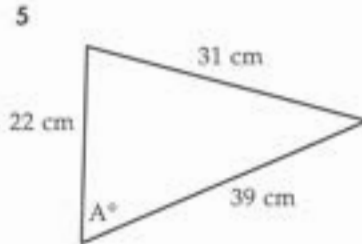
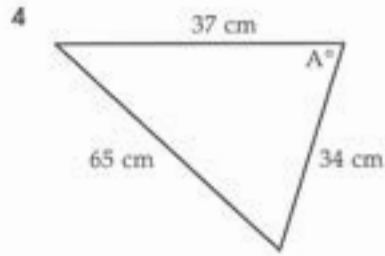
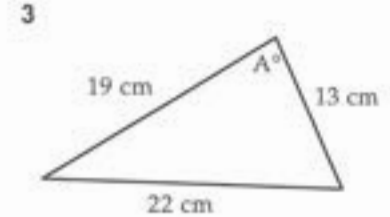
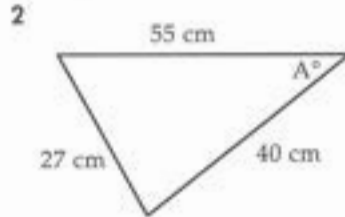
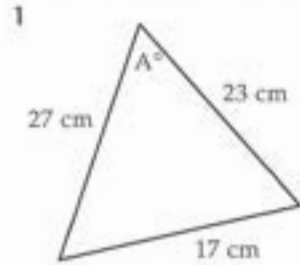
Use the cosine rule to find the size of the **obtuse** angle A to the nearest degree.



Use the cosine rule to find the size of the **obtuse** angle A to the nearest degree.



C Use the cosine rule to find the size of angle A to the nearest minute.



A

The cosine rule states that in any triangle ABC:

$$a^2 = b^2 + c^2 - 2bc \cos A$$

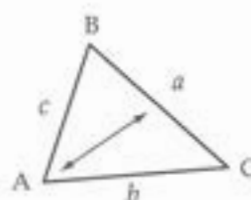
$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

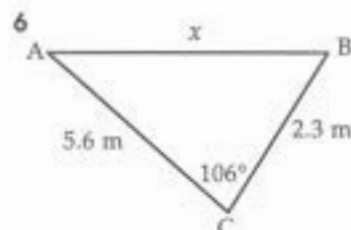
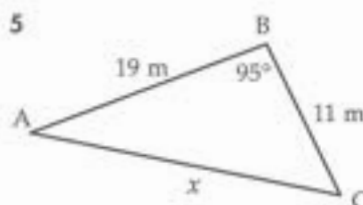
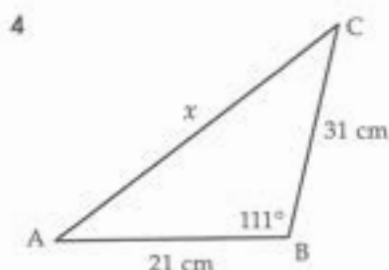
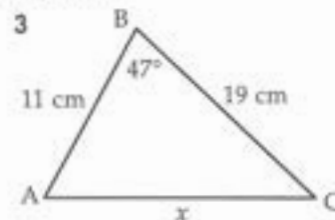
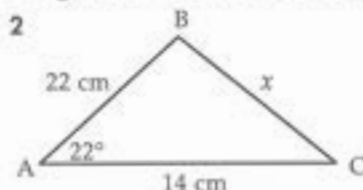
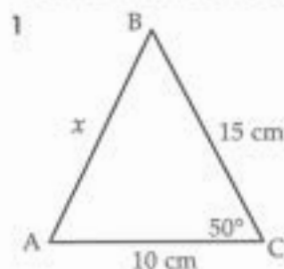
$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

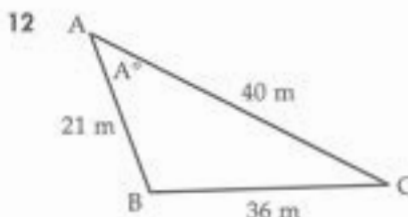
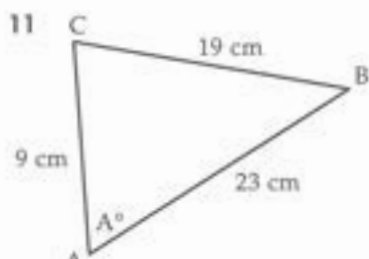
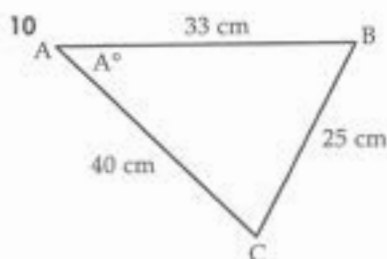
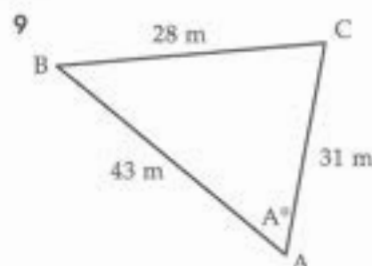
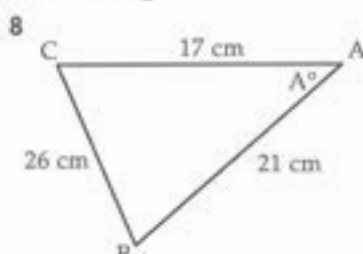
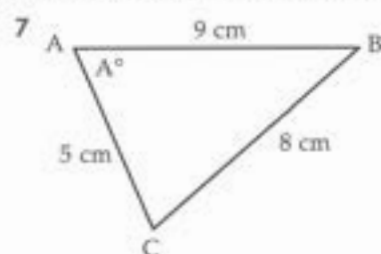
$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$



Use the cosine rule to calculate the length of side x correct to 1 decimal place.

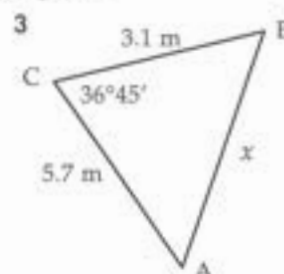
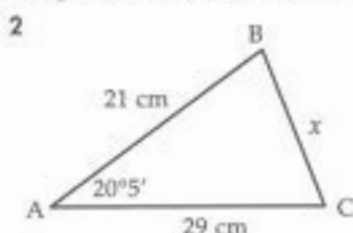
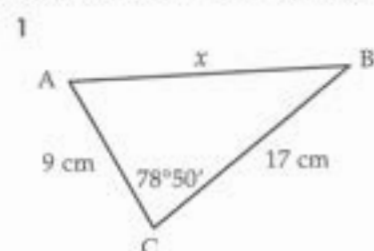


Use the cosine rule to calculate the size of angle A to the nearest degree.

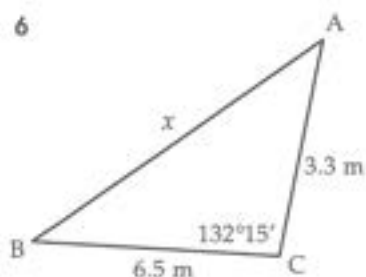
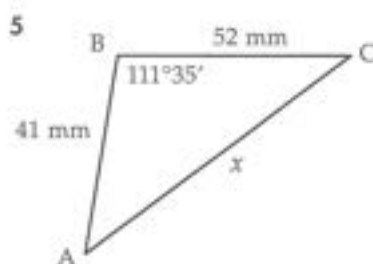
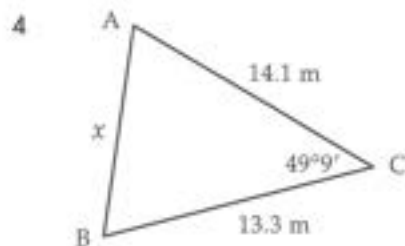


B

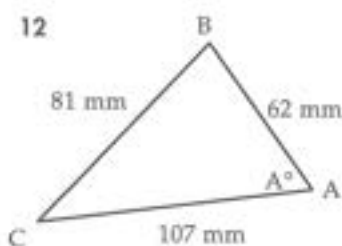
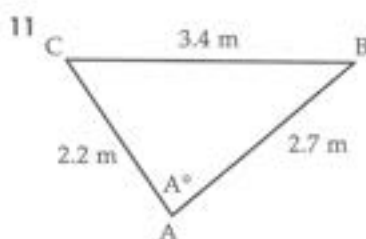
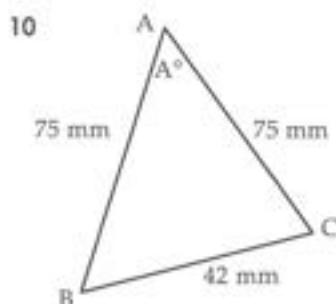
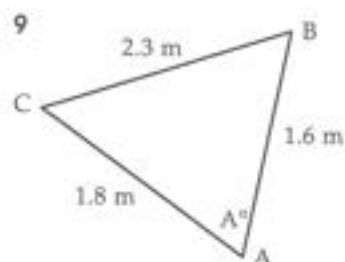
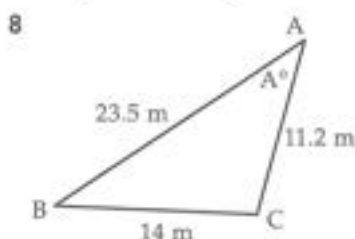
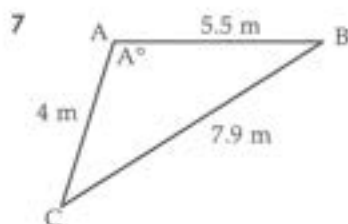
Use the cosine rule to calculate the length of side x correct to 1 decimal place.



Use the cosine rule to calculate the length of side x correct to 1 decimal place.



Use the cosine rule to find the size of angle A in degrees and minutes.

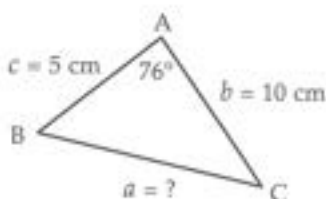


C

EXAMPLE:

In the triangle ABC, $A = 76^\circ$,
 $b = 10$ cm and $c = 5$ cm.
 Calculate the length of side a
 correct to 1 decimal place.

Use the cosine rule because you are given the lengths of
 2 sides and the included angle. Draw and label a sketch.



$$\begin{aligned} a^2 &= b^2 + c^2 - 2bc \cos A \\ &= 10^2 + 5^2 - 2(10)(5) \cos 76^\circ \\ a &= 10.0 \text{ cm} \end{aligned}$$

For each question, draw a triangle ABC and label it with the given information. Then calculate the length of side a correct to 1 decimal place.

1 $A = 56^\circ$, $b = 6$ cm and $c = 11$ cm

2 $A = 28^\circ$, $b = 33$ cm and $c = 41$ cm

3 $A = 23^\circ 35'$, $b = 19.2$ m and $c = 11.4$ m

4 $A = 77^\circ 15'$, $b = 23.4$ m and $c = 37.6$ m

Find the size of angle A in degrees and minutes.

5 $a = 15$ cm, $b = 13$ cm and $c = 18$ cm

6 $a = 112$ cm, $b = 32$ cm and $c = 90$ cm

7 $b = 14.1$ m, $c = 9.7$ m and $a = 19.6$ m

8 $c = 1.6$ m, $a = 3.3$ m and $b = 3.5$ m

Mixed questions: the sine and cosine rules 10.1

A

The **sine rule** states that in any triangle ABC:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

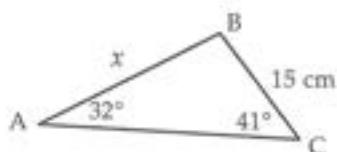
The **cosine rule** states that in any triangle ABC:

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

EXAMPLE: Find the length of side x correct to 1 decimal place.

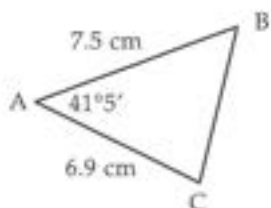


$$\frac{x}{\sin 41^\circ} = \frac{15}{\sin 32^\circ}$$

$$x = \frac{15 \times \sin 41^\circ}{\sin 32^\circ}$$

$$= 18.6 \text{ cm}$$

EXAMPLE: Find the length of side a correct to 2 decimal places.

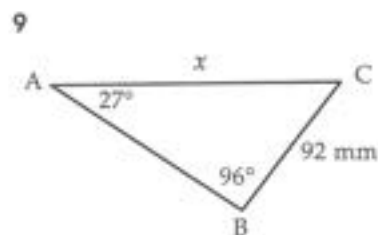
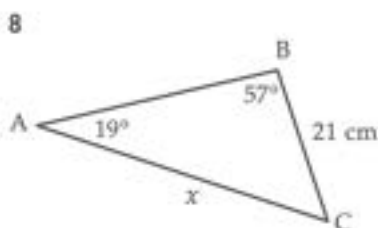
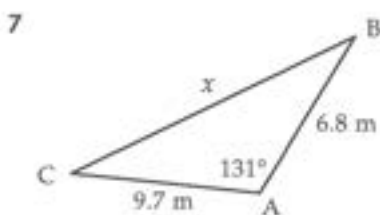
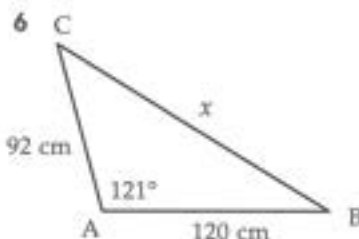
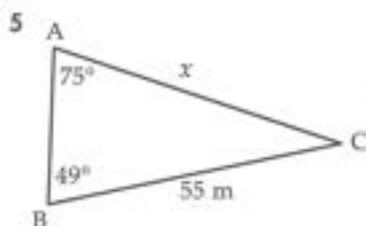
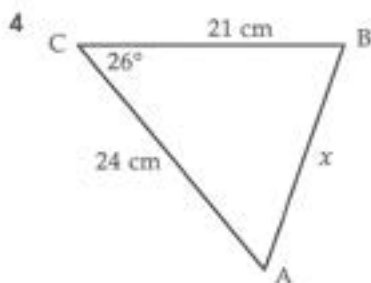
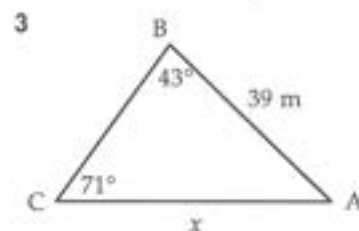
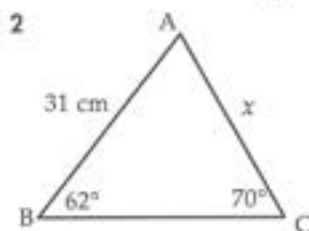
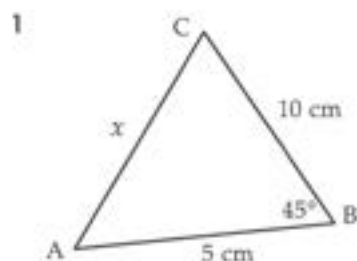


$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$a^2 = (6.9)^2 + (7.5)^2 - 2(6.9)(7.5) \cos 41^\circ 5'$$

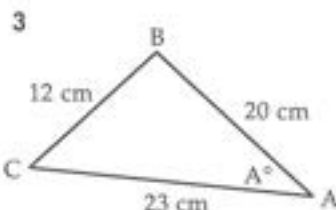
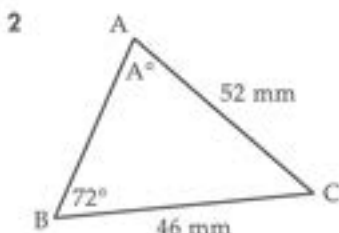
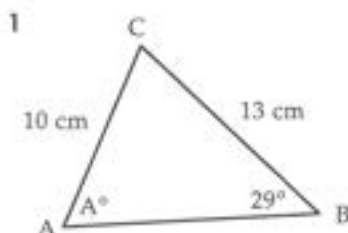
$$a = 5.08 \text{ cm}$$

Use the sine rule or the cosine rule to calculate the length of side x correct to 1 decimal place.

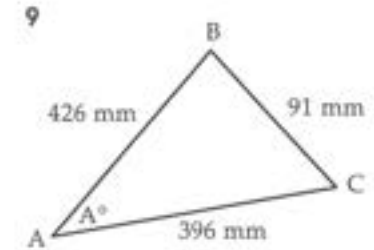
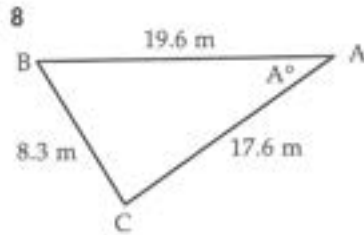
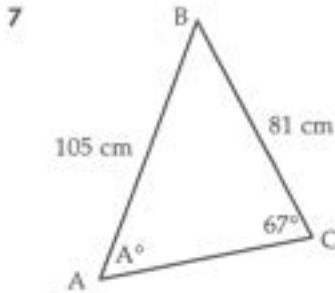
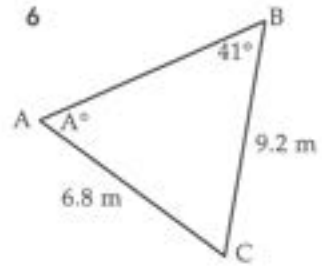
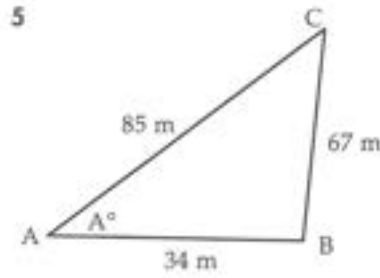
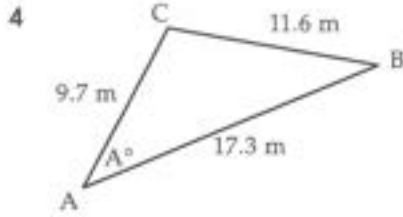


B

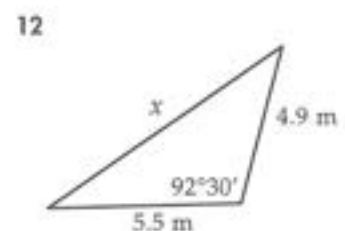
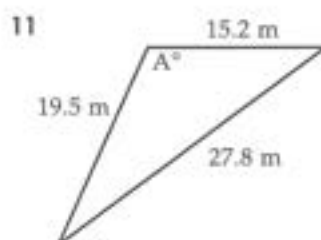
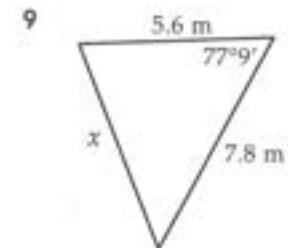
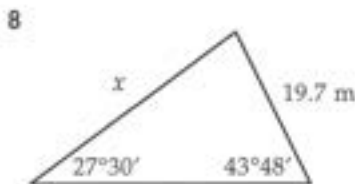
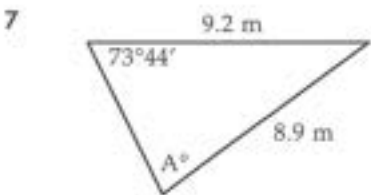
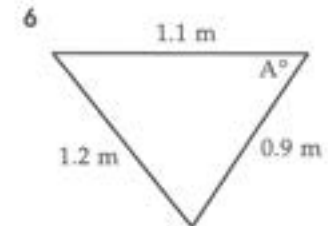
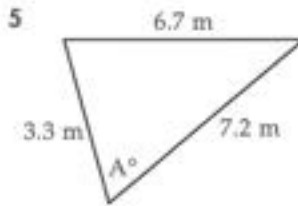
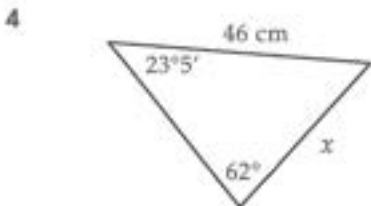
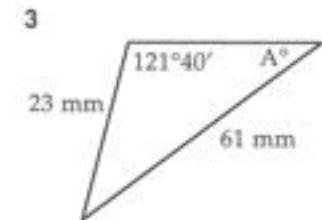
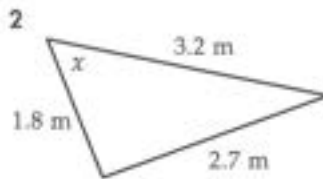
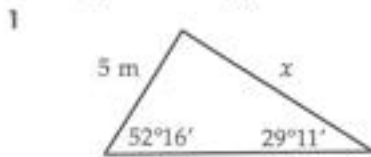
Use the sine rule or the cosine rule to find the size of the **acute** angle A to the nearest degree.



Use the sine rule or the cosine rule to find the size of the acute angle A to the nearest degree.

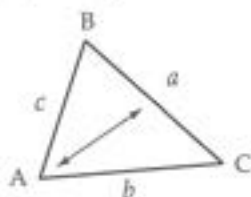


C Use the sine rule or the cosine rule to calculate the length of side x correct to 1 decimal place and the angle A in degrees and minutes.



A

For any triangle ABC:



sine rule

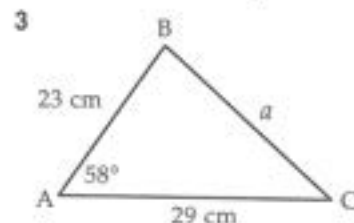
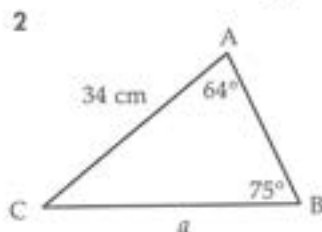
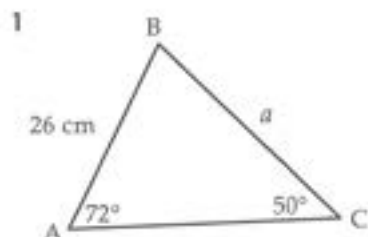
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

cosine rule

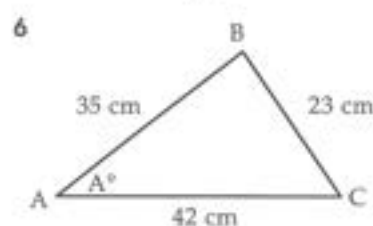
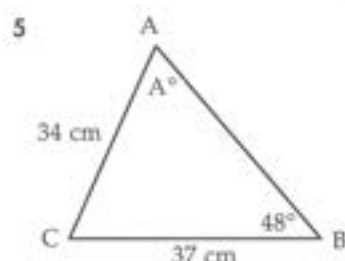
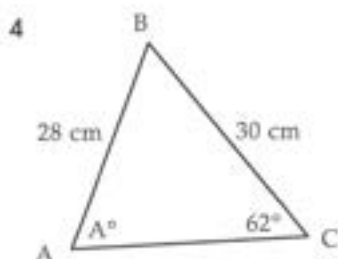
$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

Use the sine rule or the cosine rule to calculate the length of side a correct to 1 decimal place.



Use the sine rule or the cosine rule to calculate the size of angle A to the nearest degree.

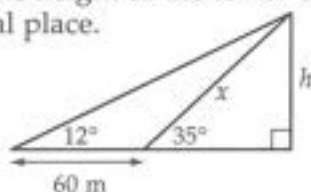


B

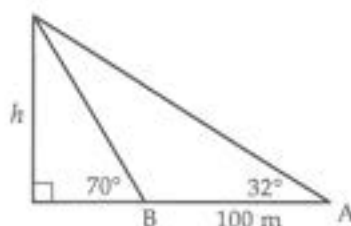
- 1** Nathan observes the top of a tower at an angle of elevation of 12° . He walks 60 metres towards the tower and finds the angle of elevation is now 35° .

a Show that: $x = \frac{60 \sin 12^\circ}{\sin 23^\circ}$

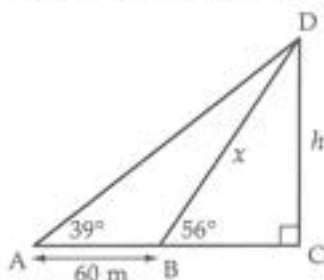
- b** Calculate the height of the tower correct to 1 decimal place.



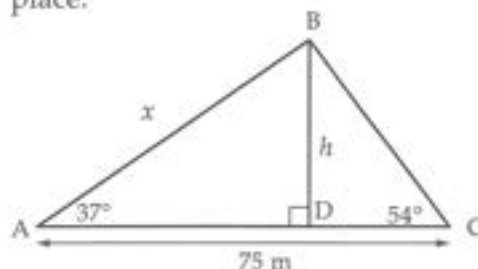
- 3** The two points A and B are directly in front of a tree. From these points, Bev measured the angles of elevation of the top of the tree as 32° and 70° respectively. A and B are 100 metres apart. Calculate the height of the tree to the nearest metre.



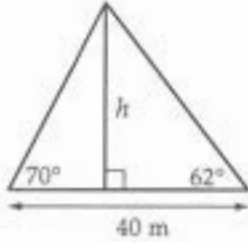
- 2** Calculate the length of BD and then find the length of CD correct to 1 decimal place.



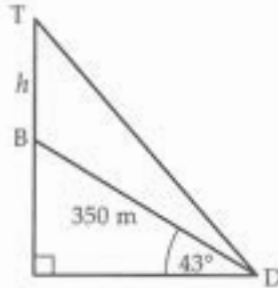
- 4** Calculate the length of AB and then find the perpendicular height BD correct to 1 decimal place.



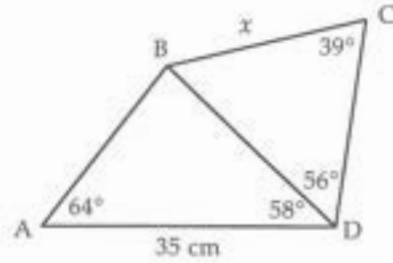
- 5 From two different points, Ruth measured the angles of elevation of the top of a tower as 70° and 62° . The points are on directly opposite sides of the tower and 40 metres apart. Calculate the height of the tower correct to 1 decimal place.



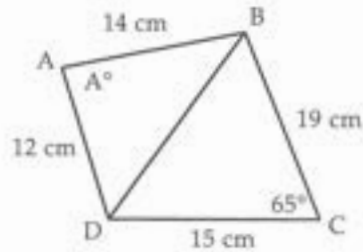
- 6 Darren is 350 metres from the base of a tower that is on a mountain. He observed the top and base of the tower at angles of elevation of 57° and 43° respectively. Calculate the height of the tower to the nearest metre.



- 7 Find the length of BD and then calculate the length of BC correct to 1 decimal place.



- 8 ABCD is a quadrilateral. Calculate the length of the diagonal BD correct to 1 decimal place and then find the size of angle A to the nearest degree.



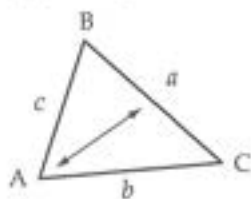
C Use the information in each question to draw a diagram. Label the diagram with the given information and then calculate the answer.

- Dana and Josh are standing 85 metres apart. They observe a glider at angles of elevation of 60° and 44° respectively. Calculate the distance between Dana and the glider and then find the height of the glider correct to 1 decimal place.
- A light plane is flying at an altitude of 1000 metres. Kamal is standing on the ground. He measured the angle of elevation of the plane to be 80° . Two minutes later, Kamal observed the plane at an angle of elevation of 15° . How far has the plane travelled in this time? (Answer to the nearest metre.)
- A step ladder with two unequal legs leans against a wall. The legs make angles of 65° and 55° with the ground and the top of the ladder reaches 120 cm up the wall. Calculate the length of each leg of the ladder to the nearest centimetre.
- A man walking along a straight, level road observes two mountain peaks in the directions 25° and 40° to the line of the road. He walks 1200 metres further and the mountain peaks are in line at an angle of 60° with the line of the road. Calculate the distance between the two mountain peaks.
- In a triangle ABC, $A = 33^\circ 27'$, $B = 42^\circ 15'$ and $a = 30$ cm. Calculate the length of the perpendicular from A to BC.
- In a quadrilateral, an angle measuring 73° is between two sides measuring 18 cm and 15 cm. The other two sides measure 16 cm and 21 cm. Calculate the size of the angle between these two sides to the nearest degree. (Hint: draw a diagonal.)

Review: the sine rule and the cosine rule 12.1

A

For any triangle ABC:



sine rule

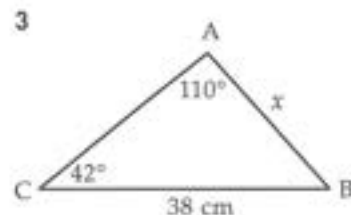
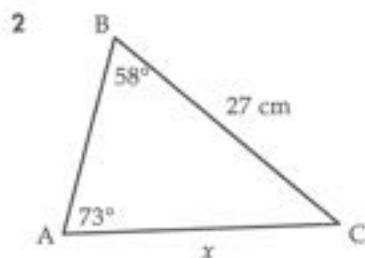
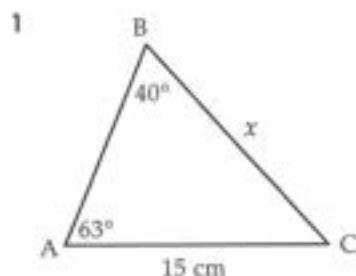
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

cosine rule

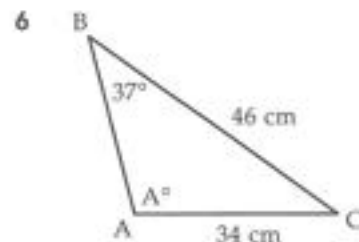
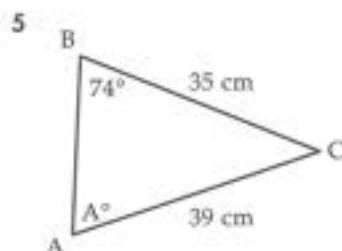
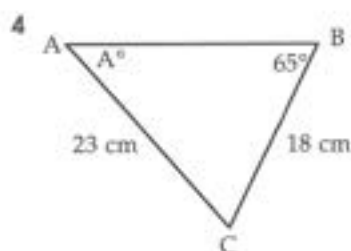
$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

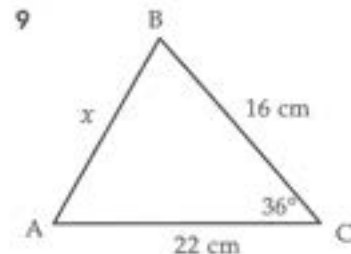
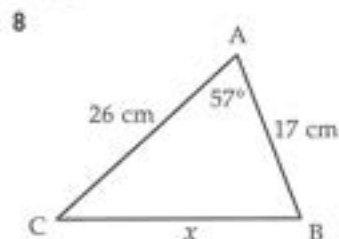
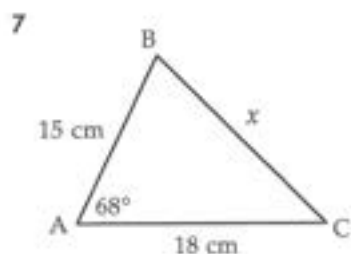
Use the sine rule to calculate the length of side x correct to 1 decimal place.



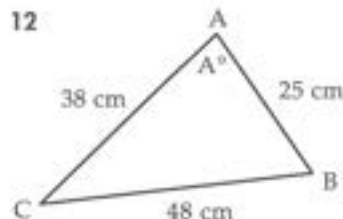
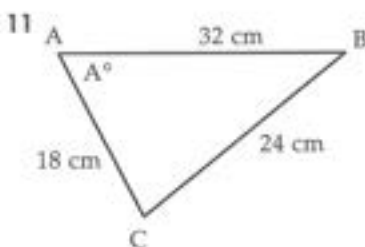
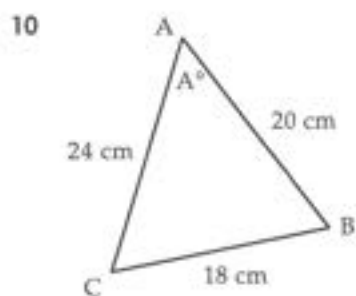
Use the sine rule to calculate the size of angle A to the nearest degree.



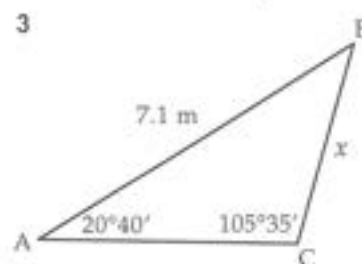
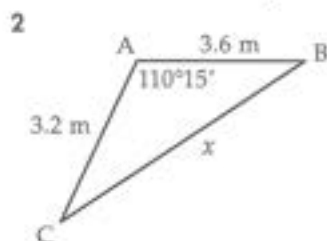
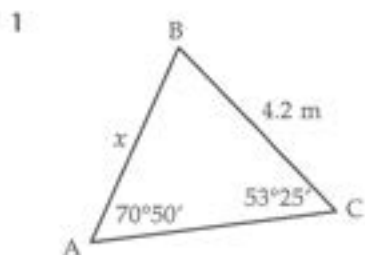
Use the cosine rule to calculate the length of side x correct to 1 decimal place.



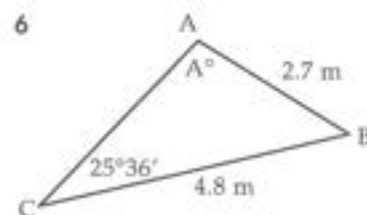
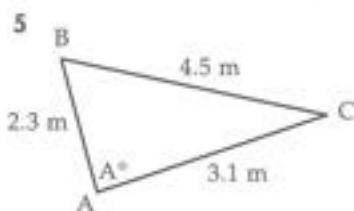
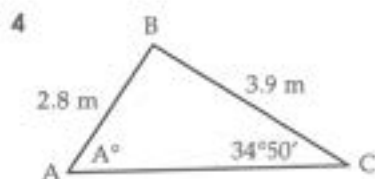
Use the cosine rule to calculate the size of angle A to the nearest degree.



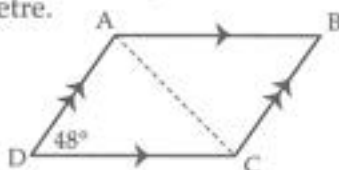
- B** Use the sine rule or the cosine rule to calculate the length of side x correct to 1 decimal place.



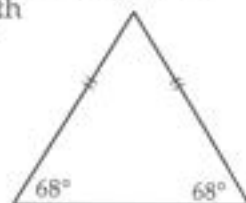
Use the sine rule or the cosine rule to find the size of angle A in degrees and minutes.



- 7 In a parallelogram ABCD, $AD = 135$ mm, $AB = 190$ mm and angle $ADC = 48^\circ$. Calculate the length of the diagonal AC to the nearest millimetre.



- 8 The two equal sides of an isosceles triangle are 165 mm long and the two equal angles are 68° . Calculate the length of the third side to the nearest millimetre.



- C** For each question, draw a triangle ABC and label it with the given information. Then calculate the length of side a correct to 1 decimal place.

1 $A = 25^\circ$, $B = 78^\circ$ and $b = 21$ cm

2 $A = 115^\circ$, $B = 43^\circ$ and $b = 17$ cm

3 $A = 123^\circ$, $b = 9$ cm and $c = 21$ cm

4 $A = 41^\circ27'$, $b = 11$ cm and $c = 19$ cm

Find the size of angle A in degrees and minutes.

5 $B = 32^\circ$, $a = 13$ cm and $b = 12$ cm

6 $a = 26$ cm, $b = 14$ cm and $c = 29$ cm

7 $C = 37^\circ20'$, $a = 86$ mm and $c = 61$ mm

8 $a = 119$ mm, $b = 44$ mm and $c = 91$ mm

- 9 A triangle has sides measuring 55 mm, 65 mm and 72 mm. Find the size of the smallest angle to the nearest degree. (Hint: The smallest angle is opposite the shortest side.)
- 10 Two straight roads leave town at an angle of 56° to each other. A car leaves town on one road and at the same time a truck leaves town on the other road. After half an hour the car has travelled 38 km and the truck has travelled 29 km. How far apart are the two vehicles? (Answer to the nearest kilometre.)
- 11 A flagpole is leaning at 7° to the vertical. Peter is standing 25 metres from the base of the pole. He measures the angle of elevation of the top of the pole as 12° . Calculate the length of the flagpole correct to 1 decimal place.
- 12 In a parallelogram ABCD, $AB = 105$ mm, the diagonal $BD = 165$ mm and angle $BAD = 100^\circ$. Find the size of the acute angle ADB between the diagonal and the side of the parallelogram, in degrees and minutes.

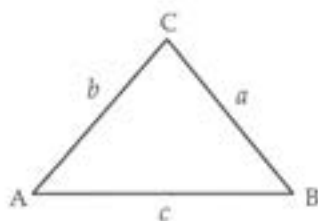
A Use your calculator to find the size of the acute angle A to the nearest degree.

- | | | | |
|--------------------|--------------------|--------------------|--------------------|
| 1 $\sin A = 0.321$ | 2 $\sin A = 0.965$ | 3 $\cos A = 0.506$ | 4 $\tan A = 3.675$ |
| 5 $\cos A = 0.035$ | 6 $\tan A = 0.832$ | 7 $\sin A = 0.801$ | 8 $\sin A = 0.671$ |

Use your calculator to evaluate each expression correct to 3 decimal places.

- | | |
|---|---|
| 9 $\frac{1}{2} \times 2 \times 4 \times \sin 23^\circ$ | 10 $\frac{1}{2} \times 5.6 \times 3.4 \times \sin 52^\circ$ |
| 11 $\frac{1}{2} \times 7.9 \times 8.1 \times \sin 17^\circ 50'$ | 12 $\frac{1}{2} \times 15.7 \times 23.1 \times \sin 23^\circ 36'$ |

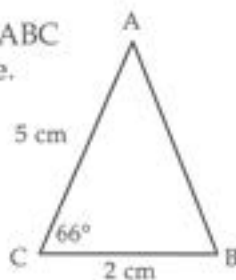
The area rule states that in any triangle ABC,
Area of $\triangle ABC = \frac{1}{2}ab\sin C$
where C is the included angle between side a
and side b.



Also: Area of $\triangle ABC = \frac{1}{2}bc\sin A$
or Area of $\triangle ABC = \frac{1}{2}ac\sin B$

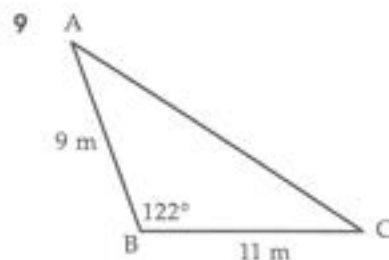
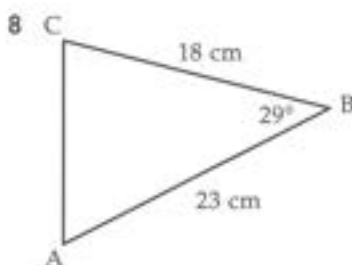
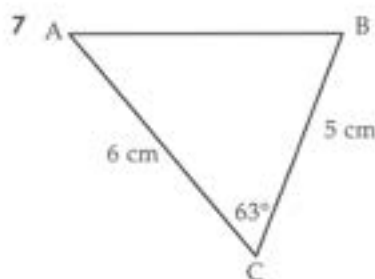
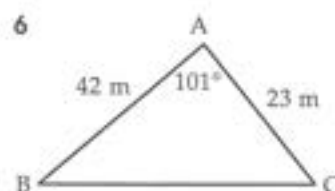
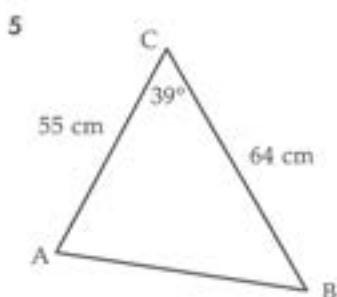
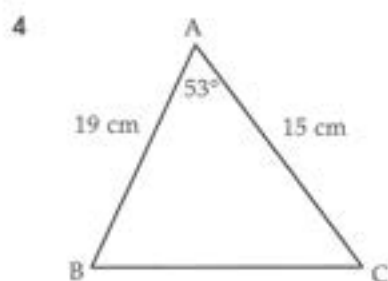
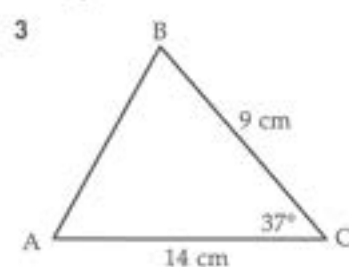
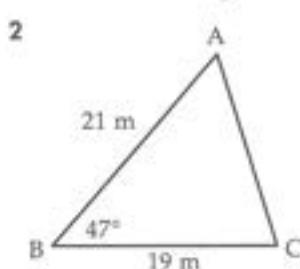
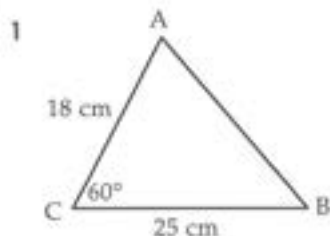
EXAMPLE:

Find the area of triangle ABC
correct to 1 decimal place.

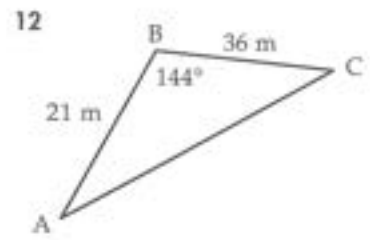
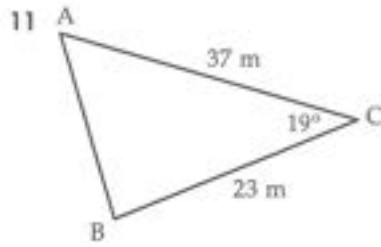
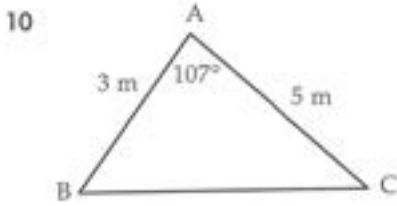


$$\begin{aligned}\text{Area of } \triangle ABC &= \frac{1}{2}ab\sin C \\ &= \frac{1}{2} \times 2 \times 5 \times \sin 66^\circ \\ &= 4.6 \text{ cm}^2\end{aligned}$$

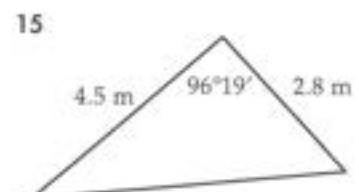
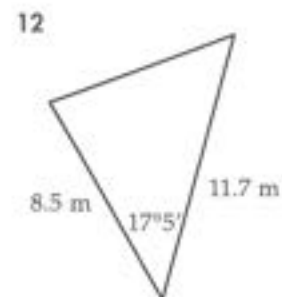
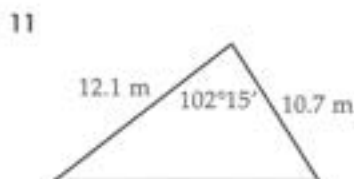
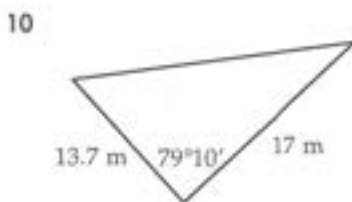
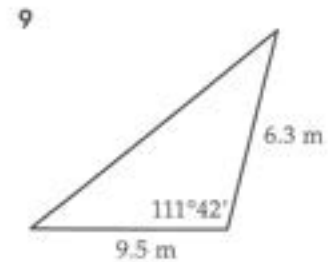
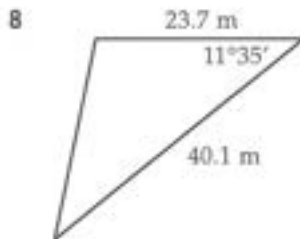
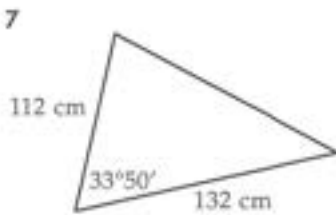
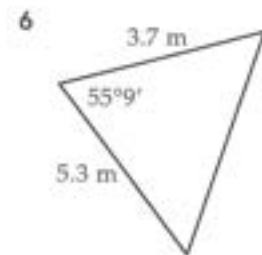
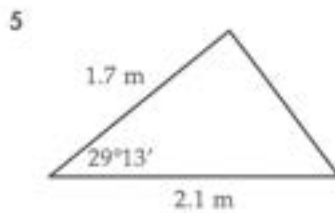
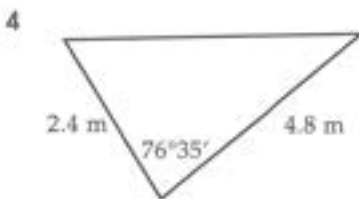
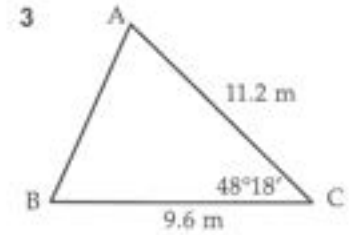
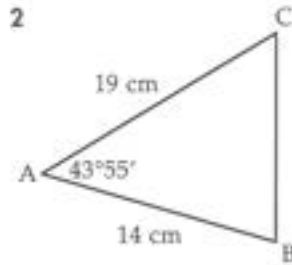
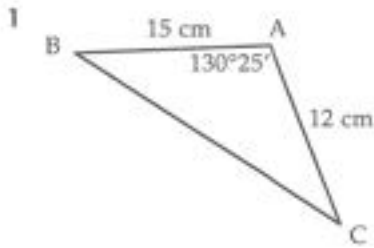
B Use the area rule to calculate the area of each triangle correct to 1 decimal place.



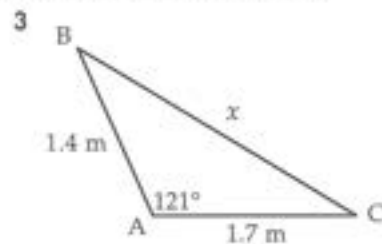
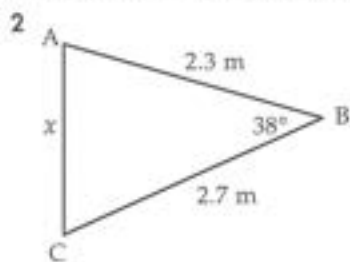
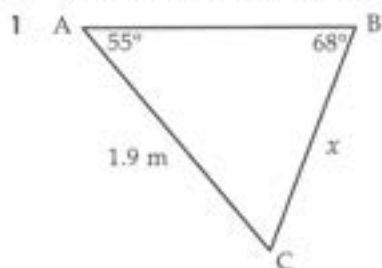
Use the area rule to calculate the area of each triangle correct to 1 decimal place.



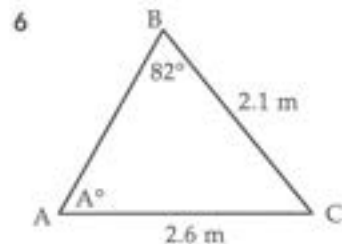
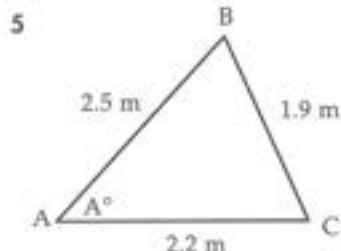
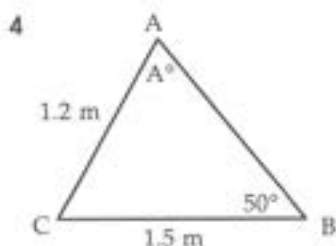
C Use the area rule to calculate the area of each triangle correct to 2 decimal places.



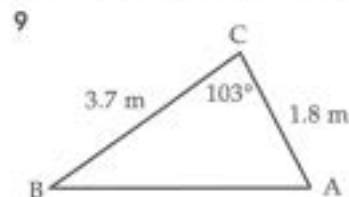
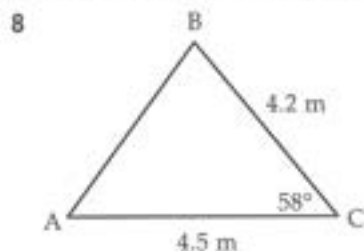
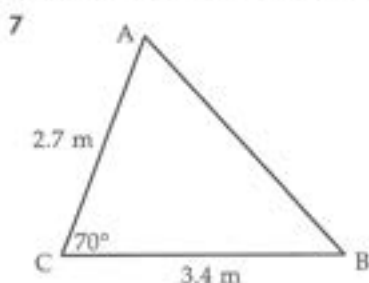
- A** Use the sine rule or the cosine rule to calculate the length of side x correct to 1 decimal place.



Use the sine rule or the cosine rule to calculate the size of angle A to the nearest degree.



Use the formula $\text{Area} = \frac{1}{2}ab\sin C$ to calculate the area of each triangle correct to 1 decimal place.

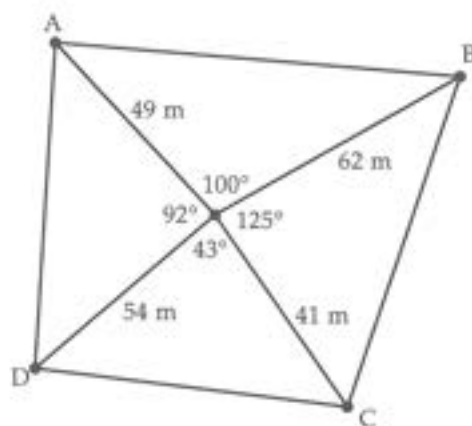


B

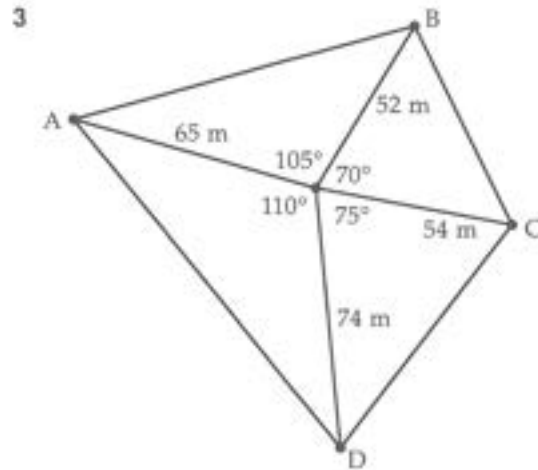
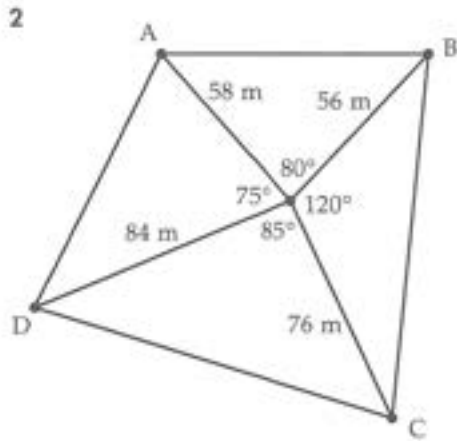
A **radial survey** is a method used to measure an irregular piece of land. It involves taking lines of sight and distances to key corner posts from a fixed position, somewhere near the centre of the field. You must be able to sight all corner posts from this one point.

- 1 A radial survey was conducted on a piece of land. The diagram shows the measurements of angles and distances from the centre point P .

- Use the cosine rule $a^2 = b^2 + c^2 - 2bc\cos A$ to calculate the distance AB correct to 1 decimal place.
- Calculate the distances BC , CD and DA correct to 1 decimal place.
- What is the perimeter of this piece of land, to the nearest metre?
- Use the formula $\text{Area} = \frac{1}{2}ab\sin C$ to calculate the area of $\triangle APB$ correct to 1 decimal place.
- Calculate the areas of the other three triangles correct to 1 decimal place.
- What is the area of this piece of land, to the nearest square metre?

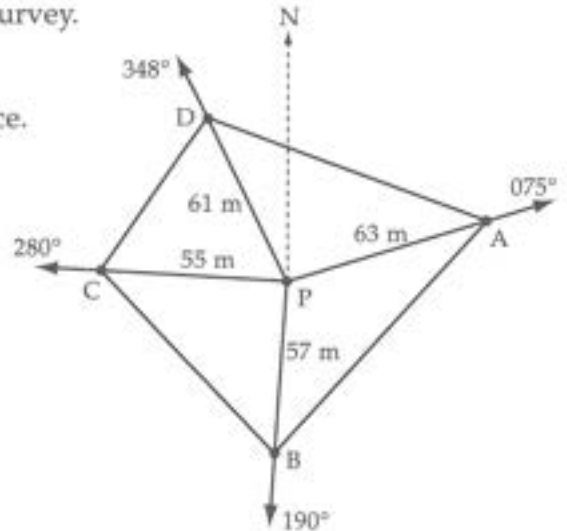


Calculate the perimeter and area of each piece of land represented below. Round your final answers to the nearest whole number.

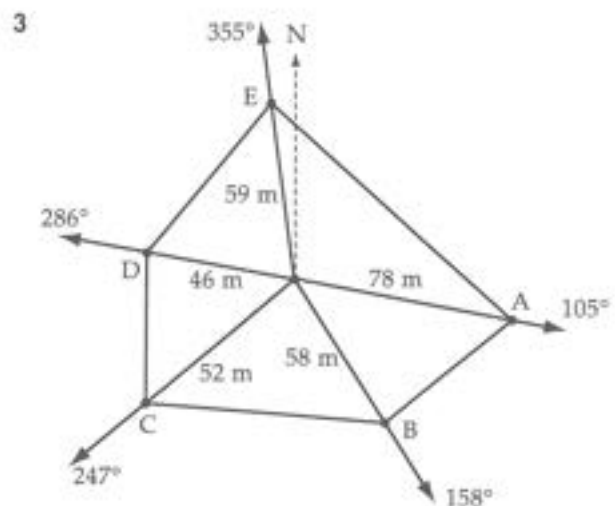
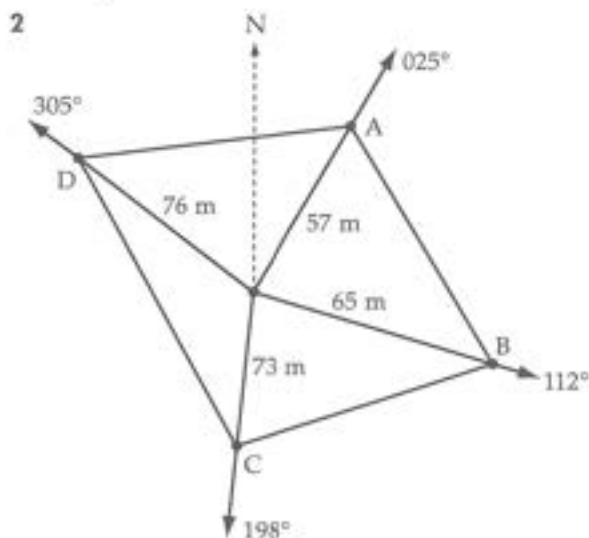


C When a compass is used to determine the directions to the corner posts, the **true bearing** (3-figure bearing) is recorded. Before doing any calculations, first determine the size of each angle.

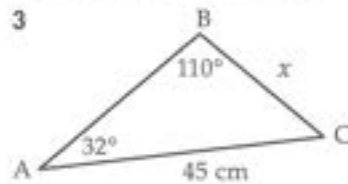
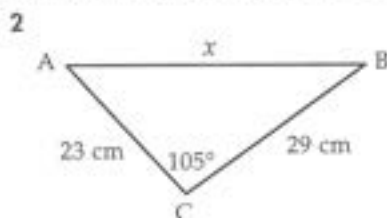
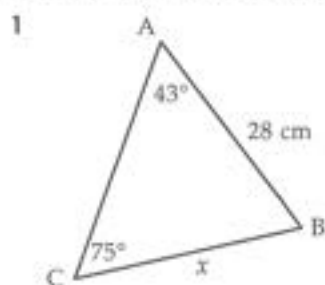
- 1 This diagram shows the measurements from a radial survey.
 - a What is the size of angle APB?
 - b What is the size of angle DPA?
 - c Calculate the distance AB, correct to 1 decimal place.
 - d What is the perimeter of this piece of land, to the nearest metre?
 - e Calculate the area of this piece of land, to the nearest square metre.



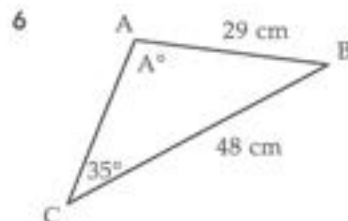
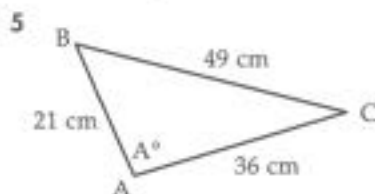
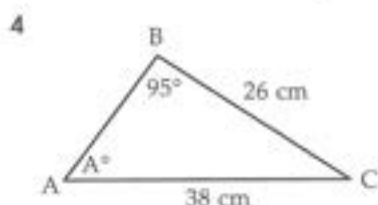
Calculate the perimeter and area of each piece of land represented below. Round your final answers to the nearest whole number.



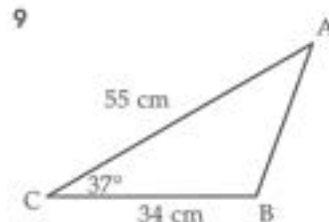
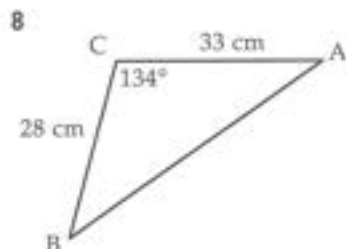
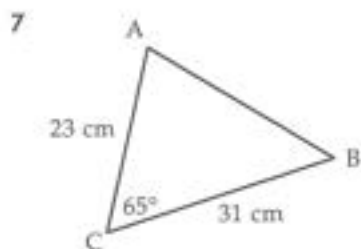
A Use the sine rule or the cosine rule to calculate the length of side x correct to 1 decimal place.



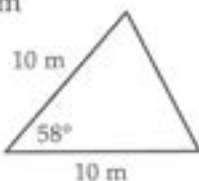
Calculate the size of angle A to the nearest degree.



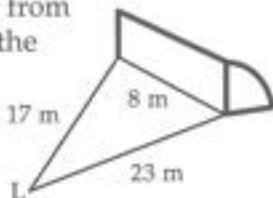
Use the formula $\text{Area} = \frac{1}{2}ab\sin C$ to calculate the area of each triangle correct to 1 decimal place.



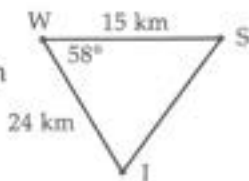
- B** 1 Two garden paths diverge from a point at an angle of 58° . Each path is 10 m long and ends at the garden fence. Calculate the area between the two paths and the fence to the nearest square metre.



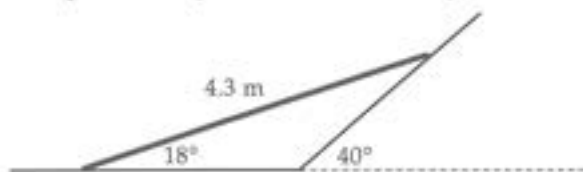
- 2 A soccer goal is 8 metres wide. Lee shoots for goal when he is 17 m from one post and 23 m from the other post. Within what angle, to the nearest degree, must he shoot to score a goal?



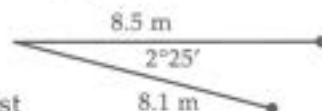
- 3 Three towns, Woodpost (W), Smithville (S) and Jamestown (J), are connected by straight roads. Woodpost to Smithville is 15 km and Woodpost to Jamestown is 24 km. The roads at Woodpost meet at an angle of 58° . How far is Smithville from Jamestown to the nearest kilometre?



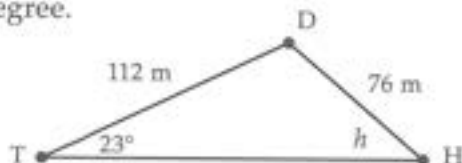
- 4 A 4.3-metre plank is placed between the ground and a ramp. The ramp is inclined at an angle of 40° to the ground. The plank makes an angle of 18° with the ground. How far up the ramp is the end of the plank?



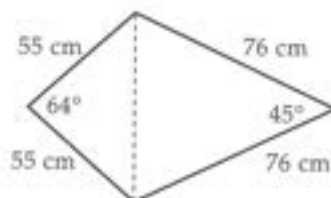
- 5 In a game of lawn bowls, Margaret is aiming to hit the jack (target bowl) 8.5 m away. Her bowl travels 8.1 m and stops $2^\circ 25'$ off the centre line. How far is Margaret's bowl from the jack? (Answer to the nearest centimetre.)



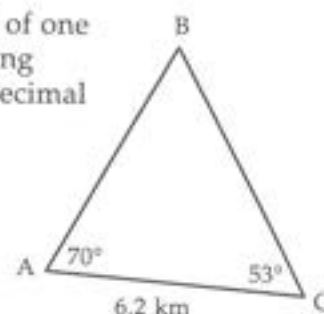
- 6 The 9th hole at Woodpost golf course has a dogleg fairway. Nathan plays his first shot from the tee (T) to a point D, 112 metres away. His ball is now 76 metres from the hole. Given that the angle $HTD = 23^\circ$, calculate the size of angle TDH to the nearest degree.



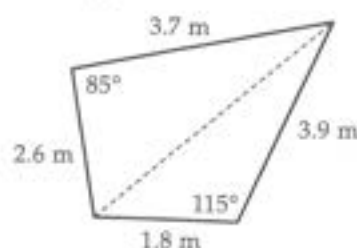
- 7 Calculate the area of the kite to the nearest square centimetre.



- 8 Calculate the length of one lap of this orienteering course correct to 1 decimal place.



- 9 Calculate the area of this quadrilateral to the nearest square metre.



C Use the information in each question to draw a diagram. Label the diagram with the given information and then calculate the answer.

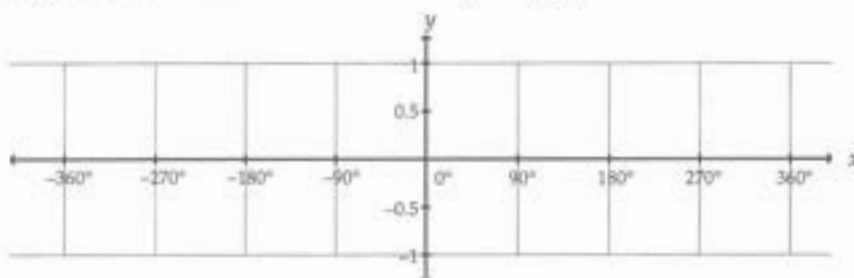
- The two equal sides in an isosceles triangle are 155 mm long and the two equal angles are 73° . Find the length of the third side of the triangle to the nearest millimetre.
- On a golf course, the distance from the tee to the hole is 400 metres in a straight line. A ball driven from the tee travels 240 metres, but is 8° off-line. How far is it from the hole? (Answer to the nearest metre.)
- Two straight roads diverge from a junction at an angle of 38° . From the junction, a car is 800 metres along one road and a motorcycle is 1250 metres along the other road. Calculate the distance between the two vehicles.
- Calculate the area of a parallelogram that has sides 9.1 m and 5.3 m and an angle of 151° . (Answer correct to 1 decimal place.)
- Three towns, A, B and C, are marked on a map. B is 28 km from A and bears $N60^\circ E$ from A. Town C bears $N17^\circ E$ from A and $N50^\circ W$ from B. Calculate the distance from A to C, correct to 1 decimal place.
- Three posts, A, B and C, are positioned so that AB is 3.3 m, BC is 4.5 m and AC is 5.1 m. Find the angle at post A between AB and AC to the nearest minute.
- An orienteering course begins at marker A and runs for 2.5 km on a bearing of 070° to marker B. The runners turn at B and run 1.8 km to marker C, which is due east of the starting point. The runners then turn and run directly to marker A. Calculate the total distance around 1 lap of this course correct to 1 decimal place.
- A kite has adjacent sides measuring 330 mm and 565 mm. The longer diagonal makes angles of 44° and 24° with the sides of the kite. Calculate the area of the kite to the nearest square millimetre.
- A ship sails from a port for 10 nautical miles in a direction 065° . It then sails in the direction 024° for 18 nautical miles. What is the distance from the ship to the port, correct to 1 decimal place?
- A bushwalker travels 8.5 km from a base camp on a bearing of 345° to point A. He then turns and walks for 9.2 km to point B, which is due west of the base camp. What is the bearing of B from A?

- A** 1 Use a calculator to complete these tables of values to two decimal places.

x	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	360°
$\sin x$													

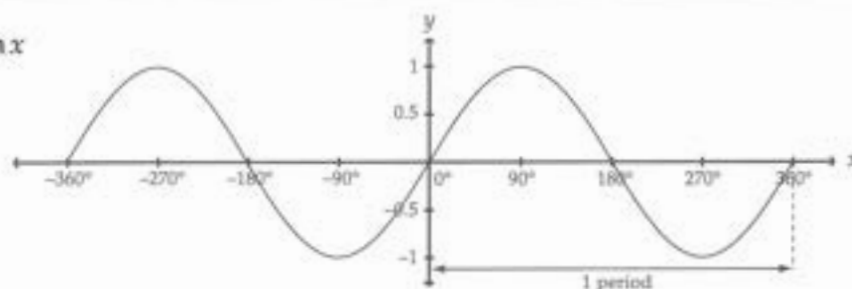
x	0°	-30°	-60°	-90°	-120°	-150°	-180°	-210°	-240°	-270°	-300°	-330°	-360°
$\sin x$													

- 2 Draw a set of axes as shown below on graph paper (2 mm grid). For the scale on the horizontal axis, have 1 cm represent 30° . For the scale on the vertical axis, have 2 cm represent 0.5 units. Plot the points from the tables above on your graph. Use a smooth curve to join the points.



B

$y = \sin x$



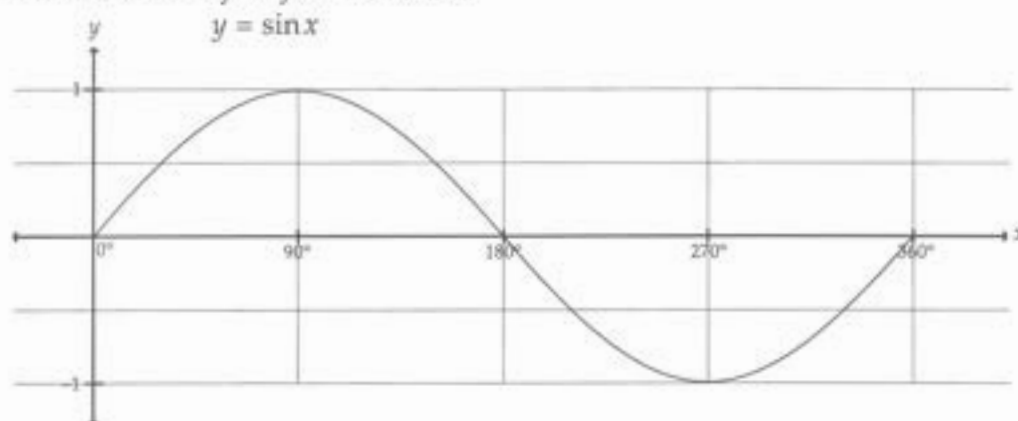
The **period** of the graph is 360° . That is, the graph repeats itself every 360° .

The **amplitude** is 1. That is:

$$\frac{\text{maximum value} - \text{minimum value}}{2}$$

The **range** = $-1 \leq y \leq 1$. That is, the maximum y value is +1 and the minimum value is -1.

Use the graph you drew in question A2 above (it should look like the graph below) to solve for x or y in the given equations. Indicate clearly on the graph where you read off each value. Then use a calculator to check the accuracy of your answers.



1 $y = \sin 90^\circ$

4 $\sin x = -1$

7 $\sin x = 0.866$

2 $y = \sin 360^\circ$

5 $\sin x = 0.5$

8 $\sin x = 0$

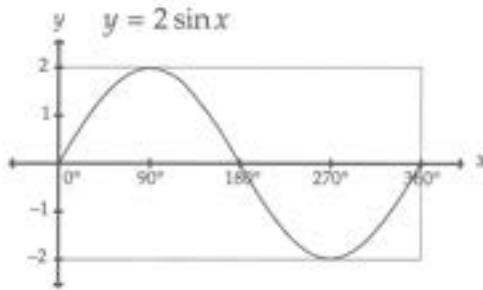
3 $y = \sin 270^\circ$

6 $\sin x = 1$

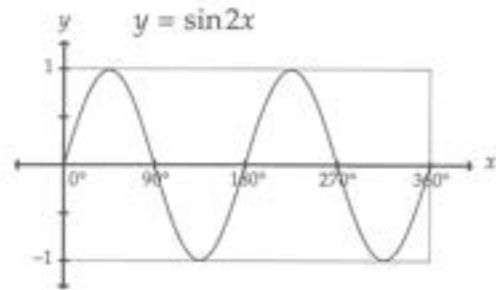
9 $\sin x = -0.5$

G

Variations of $y = \sin x$: $y = a \sin x$ and $y = \sin bx$



- The period is still 360° .
- The amplitude changes to $|a| = 2$. That is, the positive value of a .
- x -intercepts remain 0° , 180° and 360° .



- The period changes to $\frac{360^\circ}{b} = \frac{360^\circ}{2} = 180^\circ$
- The amplitude is still 1.
- The x -intercepts are 0° , 90° , 180° , 270° and 360° .

Sketch each graph for the domain $0^\circ \leq x \leq 360^\circ$.

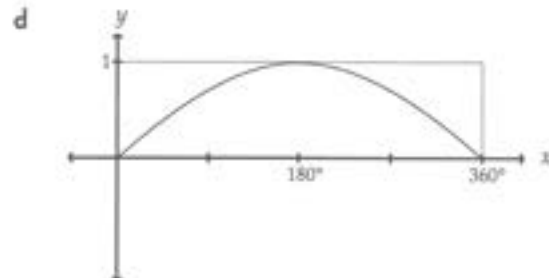
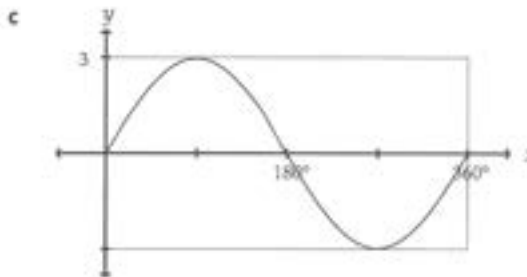
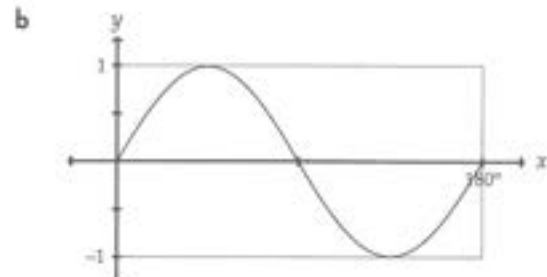
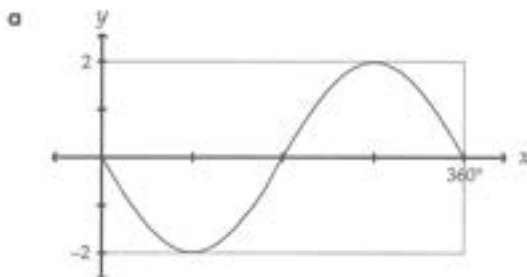
1 $y = 2 \sin x$

2 $y = \sin 2x$

3 $y = -\sin x$

4 Match each of the functions below to the correct graph.

$y = \sin 2x$, $y = \sin 3x$, $y = \sin \frac{1}{2}x$, $y = -2 \sin x$



Complete each statement.

- The maximum value of $y = 3 \sin x$ is _____.
- The period of $y = \sin x$ is _____.
- The period of $y = \frac{1}{2} \sin x$ is _____.
- The period of $y = \sin 3x$ is _____.
- The minimum value of $y = \sin 2x$ is _____.

The graph of $y = \cos x^\circ$

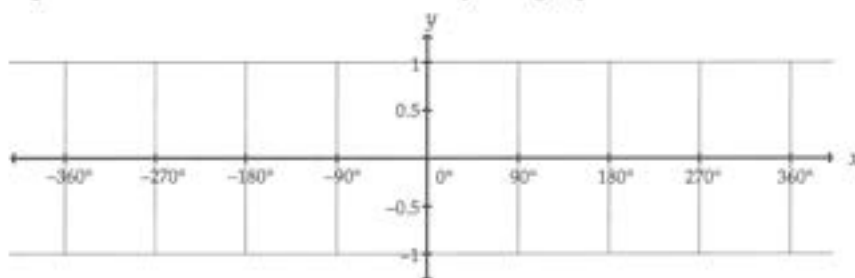
17.1

- A** 1 Use a calculator to complete these tables of values to two decimal places.

x	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	360°
$\cos x$													

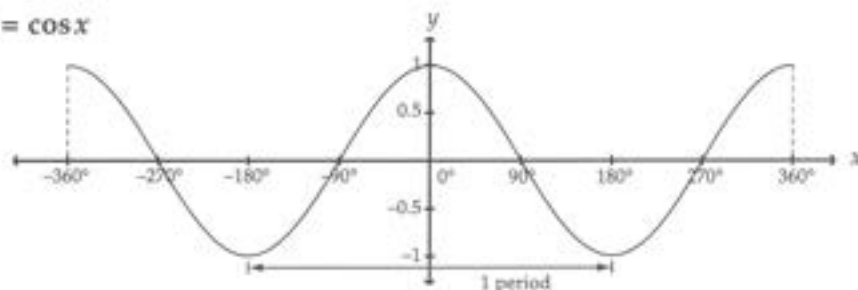
x	0°	-30°	-60°	-90°	-120°	-150°	-180°	-210°	-240°	-270°	-300°	-330°	-360°
$\cos x$													

- 2 Draw a set of axes as shown below on graph paper (2 mm grid). For the scale on the horizontal axis, have 1 cm represent 30° . For the scale on the vertical axis, have 2 cm represent 0.5 units. Plot the points from the tables above on your graph. Use a smooth curve to join the points.



B

$y = \cos x$



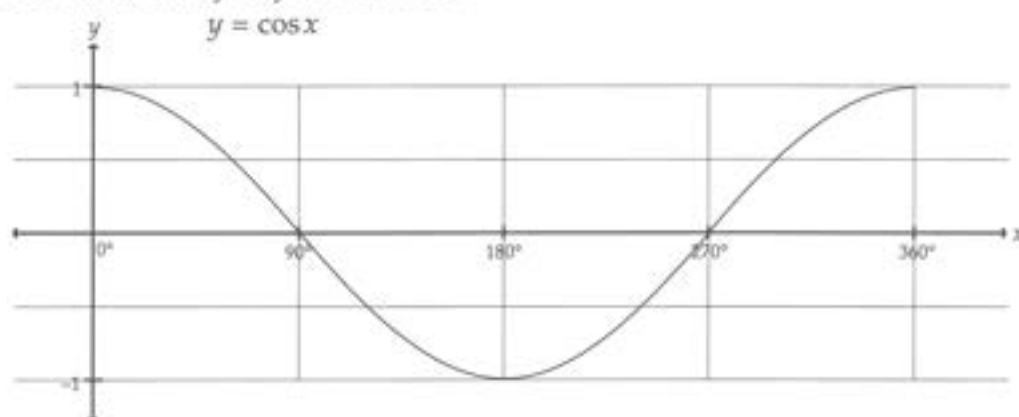
The **period** of the graph is 360° . That is, the graph repeats itself every 360° .

The **amplitude** is 1. That is:

$$\frac{\text{maximum value} - \text{minimum value}}{2}$$

The **range** = $-1 \leq y \leq 1$. That is, the maximum y value is +1 and the minimum value is -1.

Use the graph you drew in question A2 above (it should look like the graph below) to solve for x or y in the given equations. Indicate clearly on the graph where you read off each value. Then use a calculator to check the accuracy of your answers.



1 $y = \cos 0^\circ$

4 $\cos x = 1$

7 $y = \cos 300^\circ$

2 $y = \cos 90^\circ$

5 $\cos x = 0.5$

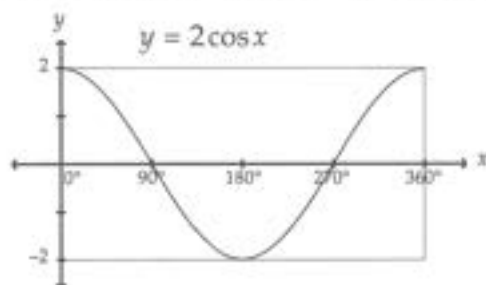
8 $y = \cos 120^\circ$

3 $y = \cos 270^\circ$

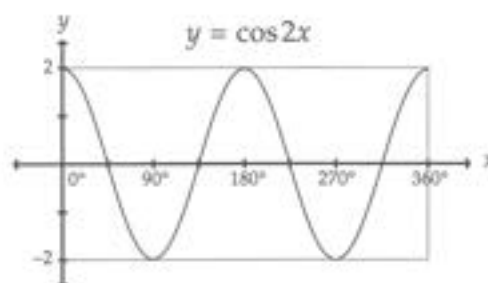
6 $\cos x = -1$

9 $y = \cos 60^\circ$

C

Variations of $y = \cos x$: $y = a \cos x$ and $y = \cos bx$ 

- The period is still 360° .
- The amplitude changes to $|a| = 2$. That is, the positive value of a .
- The x -intercepts remain 90° and 270° .

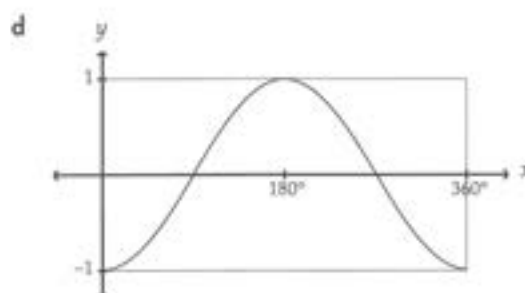
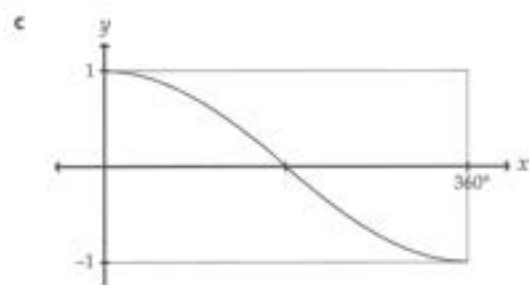
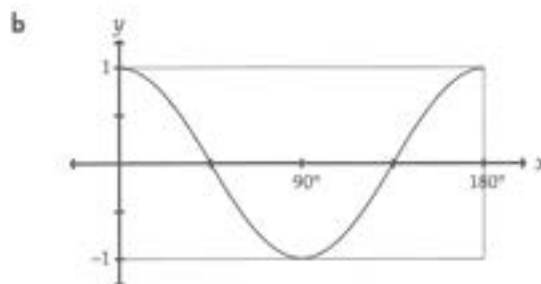
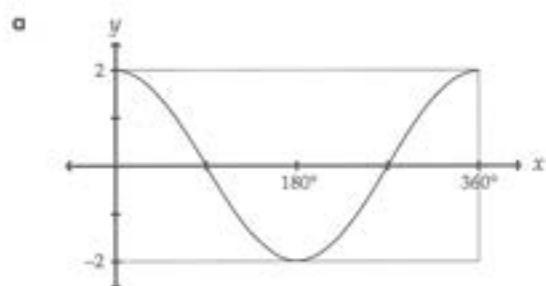


- The period changes to $\frac{360^\circ}{b} = \frac{360^\circ}{2} = 180^\circ$
- The amplitude is still 1.
- The x -intercepts are $45^\circ, 135^\circ, 225^\circ$ and 315° .

Fill in the table below and then sketch each graph for the domain $0^\circ \leq x \leq 360^\circ$.

	0°	30°	45°	60°	90°	120°	135°	180°	225°	270°	360°
1 $y = \cos x$											
2 $y = 2 \cos x$											
3 $y = \cos 2x$											
4 $y = -\cos x$											

5 Match each of the functions below to the correct graph.

 $y = \cos 2x, y = 2 \cos x, y = \cos \frac{1}{2}x, y = -\cos x$ 

Complete each statement.

- 6 The maximum value of $y = \cos x$ is ____.
- 7 The period of $y = 2 \cos x$ is ____.

- 8 The amplitude of $y = 3 \cos x$ is ____.
- 9 The period of $y = \cos 2x$ is ____.

The graph of $y = \tan x^\circ$

18.1

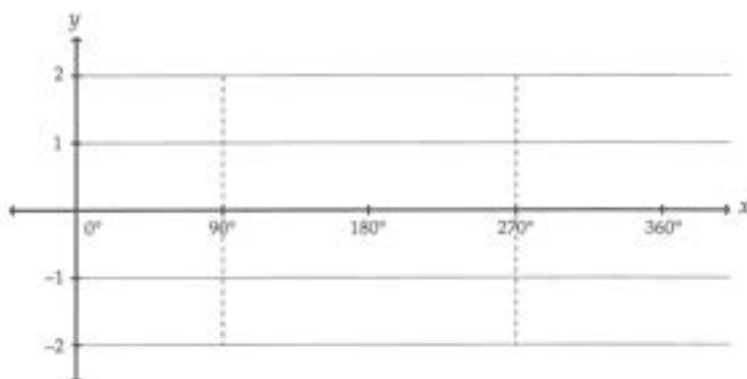
- A** 1 Use a calculator to complete these tables of values to two decimal places.

x	0°	30°	45°	60°	90°	120°	135°	150°	180°
$\tan x$									

x	210°	225°	240°	270°	300°	315°	330°	360°
$\tan x$								

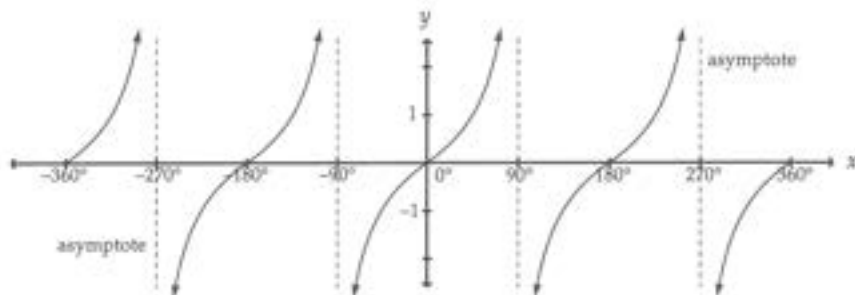
- 2 Draw a set of axes as shown below on graph paper (2 mm grid). For the scale on the horizontal axis, have 1 cm represent 45° . For the scale on the vertical axis, have 2 cm represent 0.5 units.

Plot the points from the tables above on your graph. Use smooth curves to join the points. Note that there are gaps at 90° and 270° where $\tan x^\circ$ is undefined.



B

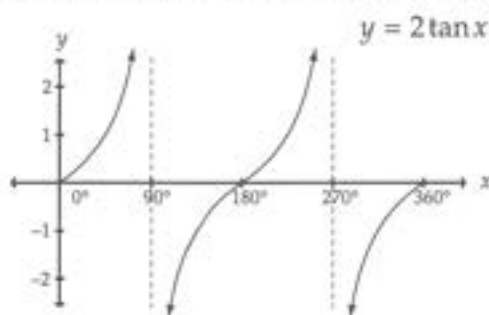
$y = \tan x$



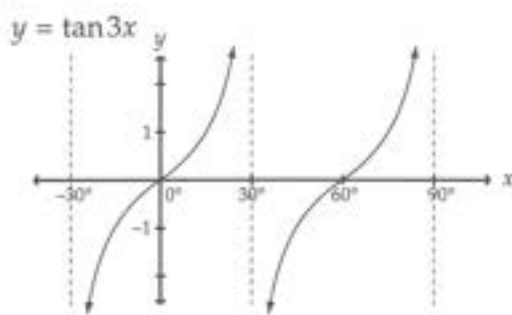
Use the graph above to complete the sentences below.

- Unlike the graphs of $\sin x$ and $\cos x$, the graph of $y = \tan x$ is not _____.
- It is clear from this graph that $y = \tan x$ is not defined for $x = \underline{\hspace{1cm}}$, $x = \underline{\hspace{1cm}}$, $x = \underline{\hspace{1cm}}$ and $x = \underline{\hspace{1cm}}$. That is, for odd multiples of 90° .
- The period of $y = \tan x$ is _____. (The graph repeats itself every ____.)
- $y = \tan x$ has no _____ because it has no maximum and minimum values.
- The perpendicular dotted lines drawn on this graph of $y = \tan x$ at the odd multiples of 90° are called _____.
- It is important to remember that:
 - $\tan x = 0$ at _____, _____, _____, _____, _____
 - $\tan x = 1$ at _____, _____, _____, _____, _____
 - $\tan x = -1$ at _____, _____, _____, _____, _____

G

Variations of $y = \tan x$: $y = a \tan x$ and $y = \tan bx$ 

- The period is still 180° .



- The period changes to $\frac{180^\circ}{b} = \frac{180^\circ}{3} = 60^\circ$

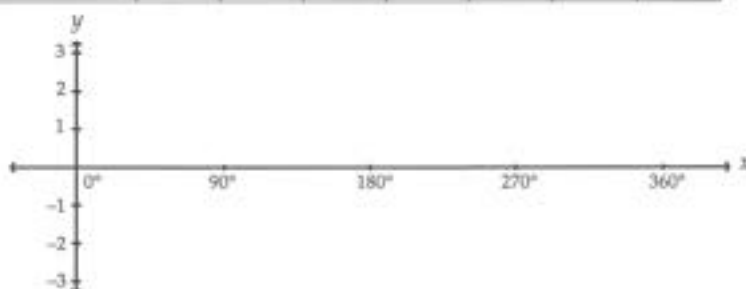
Use the graphs above to complete the sentences below.

- 1 The period of $y = 2 \tan x$ remains _____.
- 2 The asymptotes in $y = 2 \tan x$ are at _____ and _____.
- 3 The period of $y = \tan 3x$ is _____.
- 4 The asymptotes in $y = \tan 3x$ are at _____ and _____.

Fill in the table and then sketch the graph.

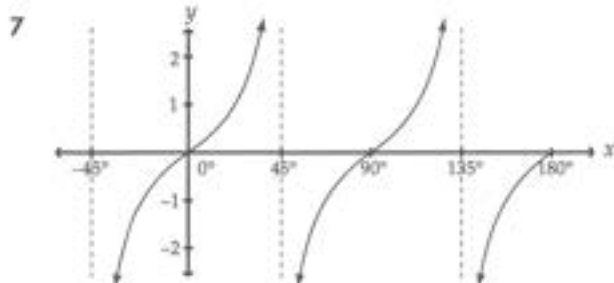
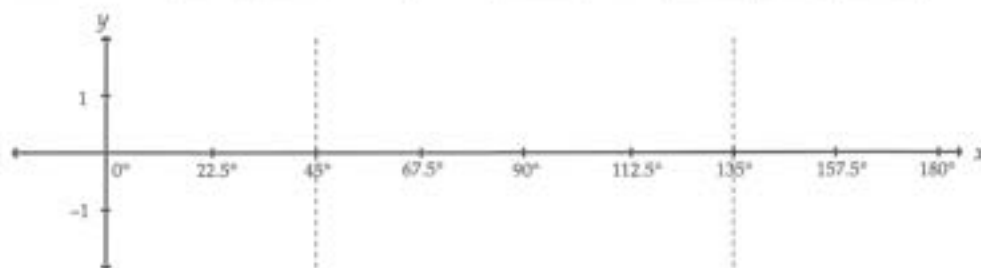
5

x	0°	45°	135°	180°	225°	315°	360°
$y = 3 \tan x$							



6

x	0°	22.5°	45°	67.5°	90°	112.5°	135°	157.5°	180°
$y = \tan 2x$									

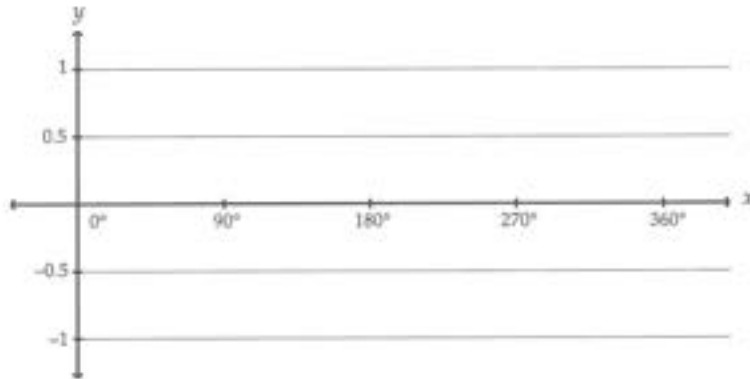


This sketch represents the graph of
 $y = \underline{\hspace{2cm}}$.

- A** 1 Use a calculator to check the table and write the missing values for a to f .

x	0°	30°	45°	60°	b°	120°	150°	180°	210°	270°	e°	360°
$\sin x$	0	0.866	0.707	a	1	0.866	c	0	d	-1	-0.5	f

- 2 Draw a set of axes as shown below on graph paper (2 mm grid). Plot the points from the tables above to draw the graph of $y = \sin x$.

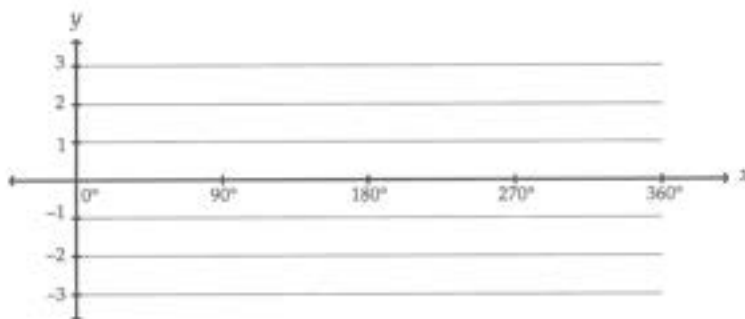


- 3 Use your graph to solve for x (to the nearest degree). Indicate clearly where you read off each value on the graph. Then use a calculator to check the accuracy of your answers.

a $\sin x = 0.707$ b $\sin x = -0.707$ c $\sin x = -0.5$ d $\sin x = 0$

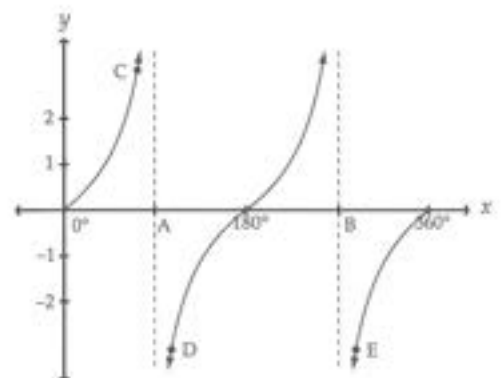
- 4 On the same set of axes, sketch the following graphs for the domain $0^\circ \leq x \leq 360^\circ$.

a $y = \frac{1}{2} \cos x$ b $y = \cos x$ c $y = 2 \cos x$ d $y = 3 \cos x$

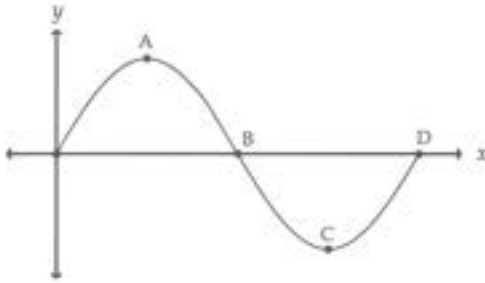


- B** 1 The diagram shows a sketch of the graph of $y = \tan x$ for $0^\circ \leq x \leq 360^\circ$. Use the graph to answer the following questions for the given domain.

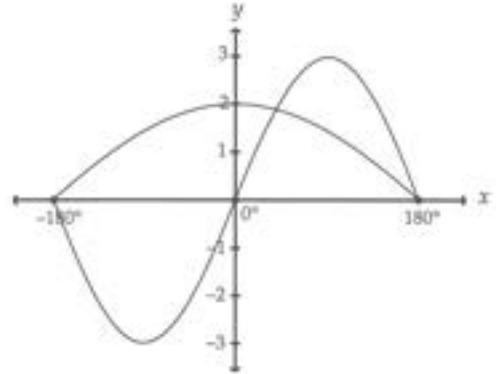
- a Give the values of x at A and B.
 b Write the range of $\tan x$.
 c C has coordinates $(73^\circ, 3.27)$. Give the x coordinates of D and E given that their y coordinate is -3.27 .
 d Given that $\tan 35^\circ = 0.7$, state the other x values for $\tan x = 0.7$.



- 2 The diagram shows a sketch graph of $y = 3 \sin x$. Write the coordinates of A, B, C and D.



- 3 The sketch shows the graphs of $y = p \sin qx$ and $y = m \cos nx$ for $-180^\circ \leq x \leq 180^\circ$.
- Find the values of p , q , m and n .
 - Find the period of each function.
 - Find the amplitude of each function.



- 1 Sketch the graphs of $y = \sin x$ and $y = \cos x$ on the same set of axes for the interval $0^\circ \leq x \leq 360^\circ$.
- Label all the x -intercepts.
 - Label all the points of intersection of $y = \sin x$ and $y = \cos x$. Use your calculator as required.
 - On the same axes, sketch the graph of $y = \sin x - \cos x$ for $0^\circ \leq x \leq 360^\circ$.
- 2 Complete the table and then sketch the graph of $y = \sec x$ for $0^\circ \leq x \leq 360^\circ$. Hint: $\sec x = \frac{1}{\cos x}$ (You may need to calculate extra values.)

x	0°	60°	90°	120°	180°	240°	270°	300°	360°
$y = \cos x$									
$y = \sec x$									

- What is the period? _____
 - What is the maximum value? _____
 - What is the minimum value? _____
- 3 Complete the table and then sketch the graph of $y = \operatorname{cosec} x$ for $0^\circ \leq x \leq 360^\circ$. Hint: $\operatorname{cosec} x = \frac{1}{\sin x}$ (You may need to calculate extra values.)

x	0°	30°	90°	150°	180°	210°	270°	330°	360°
$y = \sin x$									
$y = \operatorname{cosec} x$									

- What is the period? _____
- What is the maximum value? _____
- What is the minimum value? _____

A Use a calculator to find the size of the acute angle A to the nearest degree.

- | | | | |
|-----------------------------|----------------------------|----------------------------|-----------------------------|
| 1 $\sin A = 0.6$ | 2 $\sin A = 0.92$ | 3 $\sin A = 0.37$ | 4 $\sin A = 0.421$ |
| 5 $\cos A = 0.3$ | 6 $\cos A = 0.75$ | 7 $\cos A = 0.54$ | 8 $\cos A = 0.869$ |
| 9 $\tan A = 0.8$ | 10 $\tan A = 1.6$ | 11 $\tan A = 2.31$ | 12 $\tan A = 0.965$ |
| 13 $\sin A = 0.742$ | 14 $\cos A = 0.628$ | 15 $\tan A = 1.546$ | 16 $\cos A = 0.023$ |
| 17 $\sin A = 0.957$ | 18 $\tan A = 9.52$ | 19 $\cos A = 0.354$ | 20 $\sin A = 0.105$ |
| 21 $\sin A = \frac{12}{13}$ | 22 $\cos A = \frac{8}{17}$ | 23 $\tan A = \frac{5}{8}$ | 24 $\sin A = \frac{24}{25}$ |
| 25 $\tan A = \frac{8}{15}$ | 26 $\sin A = \frac{5}{12}$ | 27 $\tan A = \frac{7}{20}$ | 28 $\tan A = \frac{7}{8}$ |

B Find the size of the obtuse angle A to the nearest degree.

- | | | | |
|-------------------|---------------------|---------------------|---------------------|
| 1 $\sin A = 0.6$ | 2 $\sin A = 0.24$ | 3 $\sin A = 0.853$ | 4 $\sin A = 0.07$ |
| 5 $\cos A = -0.2$ | 6 $\cos A = -0.43$ | 7 $\cos A = -0.75$ | 8 $\cos A = -0.158$ |
| 9 $\tan A = -0.5$ | 10 $\tan A = -1.25$ | 11 $\tan A = -1.96$ | 12 $\tan A = -3.14$ |

EXAMPLE: Solve the equation $\sin A = 0.6$ given $0^\circ \leq A \leq 180^\circ$. Answer to the nearest degree.
 $A = 37^\circ$ or $A = 180^\circ - 37^\circ$
 $= 143^\circ$

This equation has two answers (see A1 and B1 above). Remember to give both answers when using the sine ratio. The cosine and tangent ratios give only one solution between 0° and 180° .

Solve these equations given $0^\circ \leq A \leq 180^\circ$. Answer to the nearest degree.

- | | | | |
|--------------------|--------------------|---------------------|---------------------|
| 13 $\sin A = 0.5$ | 14 $\sin A = 0.7$ | 15 $\sin A = 0.86$ | 16 $\sin A = 0.23$ |
| 17 $\cos A = 0.42$ | 18 $\cos A = -0.5$ | 19 $\cos A = -0.72$ | 20 $\cos A = 0.866$ |
| 21 $\tan A = 0.5$ | 22 $\tan A = -1.5$ | 23 $\tan A = 2.5$ | 24 $\tan A = -1.36$ |

Some equations need to be rearranged before you can solve them.

EXAMPLE: Solve the equation $\tan A + 1 = 0$ given $0^\circ \leq A \leq 180^\circ$. Answer to the nearest degree.
 rearrange: $\tan A = -1$
 $A = 135^\circ$

Solve these equations given $0^\circ \leq A \leq 180^\circ$. Answer to the nearest degree.

- | | | | |
|-----------------------|-----------------------|---------------------|-----------------------|
| 25 $\sin A - 0.4 = 0$ | 26 $\cos A - 0.5 = 0$ | 27 $\tan A - 1 = 0$ | 28 $\sin A - 0.8 = 0$ |
| 29 $\cos A + 0.3 = 0$ | 30 $\tan A + 2 = 0$ | 31 $2\sin A = 1$ | 32 $\tan A - 3 = 0$ |
| 33 $3\cos A = 2$ | 34 $2\tan A = -5$ | 35 $2\cos A = -1$ | 36 $4\sin A - 3 = 0$ |

C Use the ASTC rule to solve these equations given $0^\circ \leq A \leq 360^\circ$. Answer to the nearest degree.

- | | | | |
|----------------------|-----------------------|----------------------|-----------------------|
| 1 $\cos A = 0.5$ | 2 $\sin A = -0.5$ | 3 $\tan A = 1.5$ | 4 $\tan A = -1$ |
| 5 $\cos A = -0.45$ | 6 $\sin A = -0.76$ | 7 $\tan A = 1.6$ | 8 $\sin A = -0.36$ |
| 9 $\cos A = 0.85$ | 10 $\cos A + 0.5 = 0$ | 11 $\tan A + 3 = 0$ | 12 $\sin A + 0.8 = 0$ |
| 13 $\sin A - 1 = 0$ | 14 $\tan A - 2 = 0$ | 15 $\cos A + 1 = 0$ | 16 $4\cos A = 3$ |
| 17 $\sin A + 1 = 0$ | 18 $5\tan A = 6$ | 19 $3\sin A = 2$ | 20 $2\tan A = 1$ |
| 21 $3\cos A - 1 = 0$ | 22 $5\cos A + 3 = 0$ | 23 $2\sin A - 1 = 0$ | 24 $2\tan A + 3 = 0$ |

A Use your calculator to find out whether each statement is true or false.

$$1 \quad \tan 20^\circ = \frac{\sin 20^\circ}{\cos 20^\circ}$$

$$2 \quad \tan 112^\circ = \frac{\sin 112^\circ}{\cos 112^\circ}$$

$$3 \quad \sin^2 40^\circ + \cos^2 40^\circ = 1$$

$$4 \quad \sin^2 135^\circ + \cos^2 135^\circ = 1$$

$$5 \quad \sin(2 \times 15^\circ) = 2 \sin 15^\circ \cos 15^\circ$$

$$6 \quad \cos(2 \times 7^\circ) = \cos^2 7^\circ - \sin^2 7^\circ$$

$$7 \quad \cos(2 \times 7^\circ) = 2 \cos^2 7^\circ - 1$$

$$8 \quad \cos(2 \times 7^\circ) = 1 - 2 \sin^2 7^\circ$$

B Some important relationships exist between the trigonometric ratios.

Reciprocal relationships

$$\sin x = \frac{1}{\operatorname{cosec} x} \quad \cos x = \frac{1}{\sec x} \quad \tan x = \frac{1}{\cot x}$$

$$\operatorname{cosec} x = \frac{1}{\sin x} \quad \sec x = \frac{1}{\cos x} \quad \cot x = \frac{1}{\tan x}$$

Quotient relationships

$$\tan x = \frac{\sin x}{\cos x} \quad \cot x = \frac{\cos x}{\sin x}$$

Square relationships

$$\sin^2 x + \cos^2 x = 1$$

$$\sec^2 x = 1 + \tan^2 x$$

$$\operatorname{cosec}^2 x = 1 + \cot^2 x$$

EXAMPLE:

Prove that:

$$\sin^2 x + \tan x \cdot \cot x + \cos^2 x = 2$$

$$\text{L.H.S.} = (\sin^2 x + \cos^2 x) + \tan x \times \frac{1}{\tan x}$$

$$= 1 + 1$$

$$= 2$$

$$= \text{R.H.S.}$$

Use the relationships and method shown above to prove each statement.

$$1 \quad \frac{\cos x}{1 - \sin^2 x} = \sec x$$

$$2 \quad \frac{\sec x}{1 + \tan^2 x} = \cos x$$

$$3 \quad \frac{1 + \cot^2 x}{\operatorname{cosec} x} = \cos x$$

$$4 \quad \frac{\sin x + \cos^2 x \cdot \operatorname{cosec} x}{\sec x \cdot \cot x} = 1$$

$$5 \quad (\sin x + \cos x)^2 = 1 + 2 \sin x \cos x$$

$$6 \quad \cot x + \tan x = \operatorname{cosec} x \cdot \sec x$$

C Prove that each statement is true.

$$1 \quad \cos x(1 + \tan^2 x) = \sec x$$

$$2 \quad \frac{1 - 2 \sin^2 x}{\sin x \cdot \cos x} = \cot x - \tan x$$

$$3 \quad \frac{\sin x + \cos x}{\tan x + 1} = \cos x$$

$$4 \quad \frac{1}{(1 - \cos x)(1 + \cos x)} = \cot^2 x + 1$$

$$5 \quad \frac{\sin^2 x + \cos^2 x}{\cot x} + \frac{\sin x}{\cot x} = 2 \tan x$$

$$6 \quad (\sec^2 x - 1)(\operatorname{cosec}^2 x - 1) = 1$$

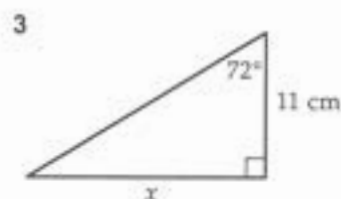
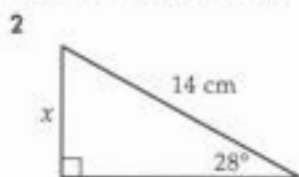
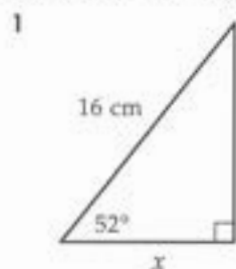
$$7 \quad 2 \tan x + 2 \cot x = \frac{2}{\sin x \cos x}$$

$$8 \quad \frac{\sec(180^\circ - x)}{\sin(180^\circ + x)} = \tan(180^\circ + x) + \cot x$$

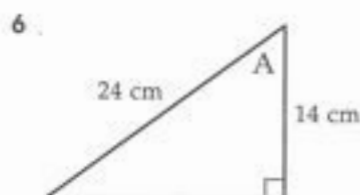
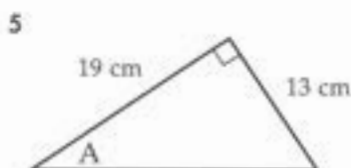
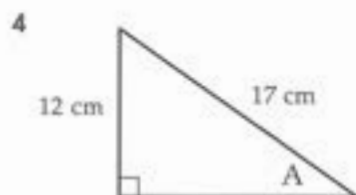
$$9 \quad \frac{\sin x}{1 - \cos x} = \frac{1 + \cos x}{\sin x}$$

$$10 \quad \sec^2 x - \sin^2 x = \cos^2 x + \tan^2 x$$

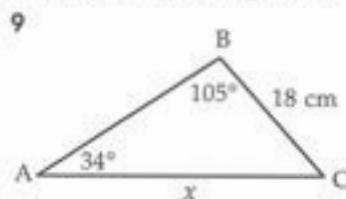
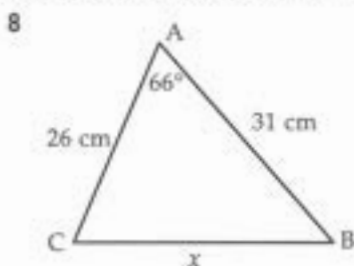
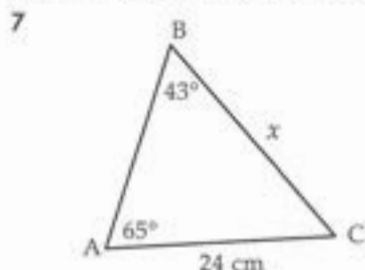
A Calculate the length of side x correct to 1 decimal place.



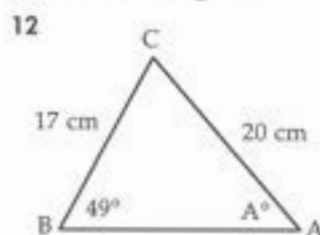
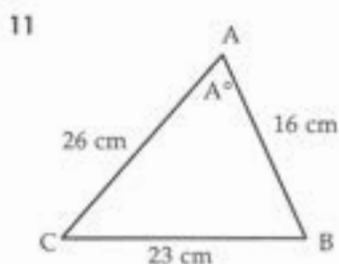
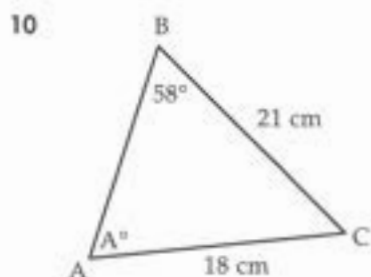
Calculate the size of angle A in degrees and minutes.



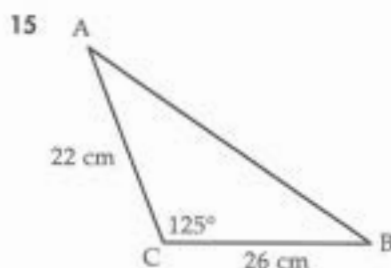
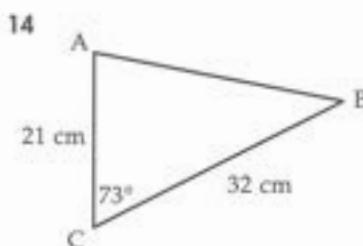
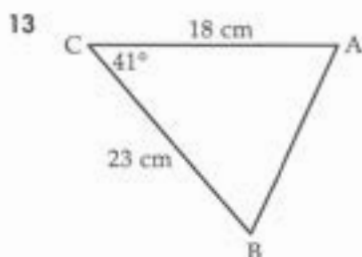
Use the sine rule or the cosine rule to calculate the length of side x correct to 1 decimal place.



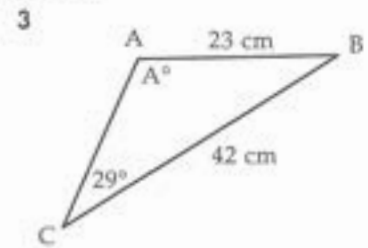
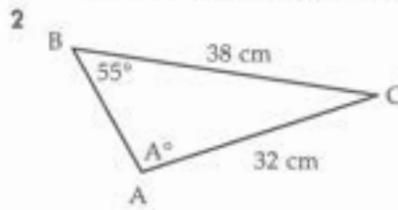
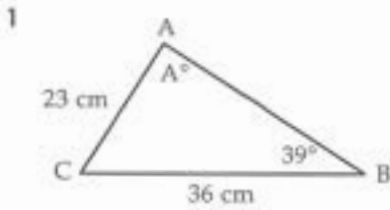
Use the sine rule or the cosine rule to calculate the size of angle A to the nearest degree.



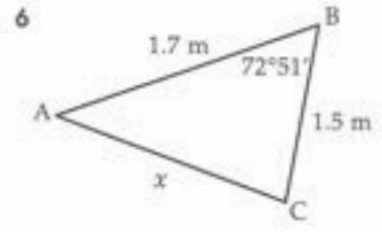
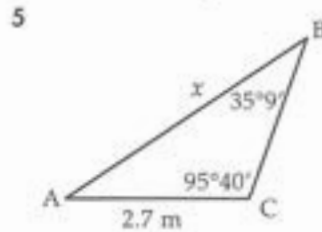
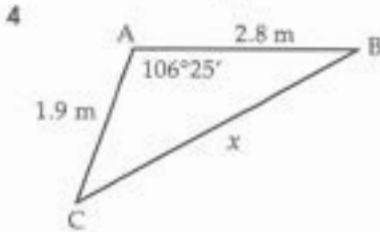
Use the formula $\text{Area} = \frac{1}{2}ab\sin C$ to calculate the area of each triangle correct to the nearest square centimetre.



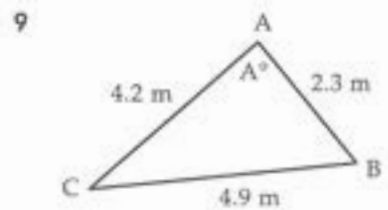
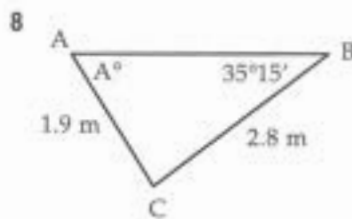
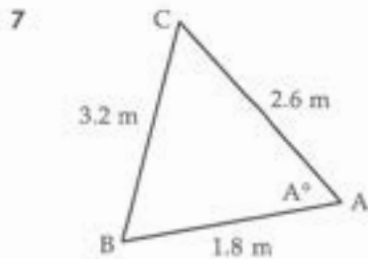
B Use the sine rule to find the size of the obtuse angle A to the nearest degree.



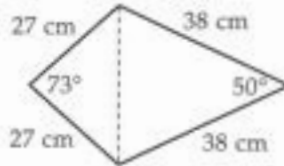
Calculate the length of side x correct to 1 decimal place.



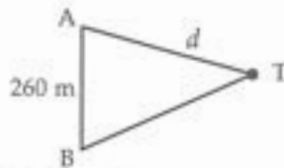
Find the size of angle A in degrees and minutes.



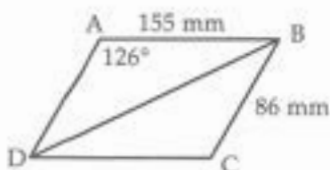
10 Calculate the area of the kite to the nearest square centimetre.



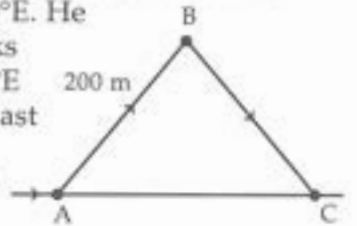
11 A tower, T, is observed from the two points A and B which are 260 metres apart. The angle TAB is 75° and angle TBA is 68° . Calculate the distance from A to the tower, to the nearest metre.



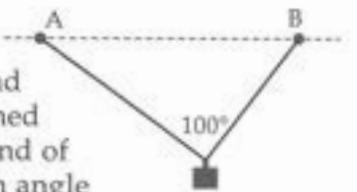
12 The parallelogram ABCD has sides measuring 86 mm and 155 mm. The angle DAB = 126° . Calculate the length of the diagonal BD to the nearest millimetre.



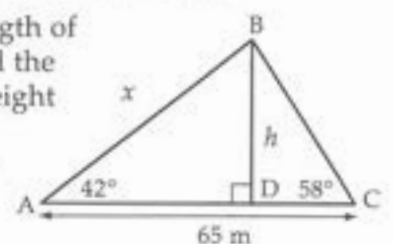
13 A surveyor walking due east turns at A to avoid a swamp and walks 200 m towards B on a bearing of $N40^\circ E$. He then turns and walks on a bearing of $S55^\circ E$ to C, which is due east of A. Calculate the distance AC correct to 1 decimal place.



14 A rope 9 metres long is attached to two level pegs, A and B. A weight is attached 4 metres from one end of the rope, forming an angle of 100° . Calculate the distance between the two pegs to the nearest millimetre.



15 Calculate the length of AB and then find the perpendicular height of BD correct to 1 decimal place.



C For each question, draw a triangle ABC and label it with the given information. Then calculate the length of side a correct to 1 decimal place.

1 $A = 58^\circ$, $B = 73^\circ$ and $b = 27$ cm

3 $A = 73^\circ$, $b = 18$ cm and $c = 14$ cm

2 $A = 43^\circ$, $B = 110^\circ$ and $b = 38$ cm

4 $A = 115^\circ$, $b = 26$ cm and $c = 22$ cm

Find the size of angle A in degrees and minutes.

5 $B = 65^\circ$, $a = 18$ cm and $b = 23$ cm

7 $a = 15$ cm, $b = 20$ cm and $c = 26$ cm

6 $C = 37^\circ$, $a = 46$ cm and $b = 34$ cm

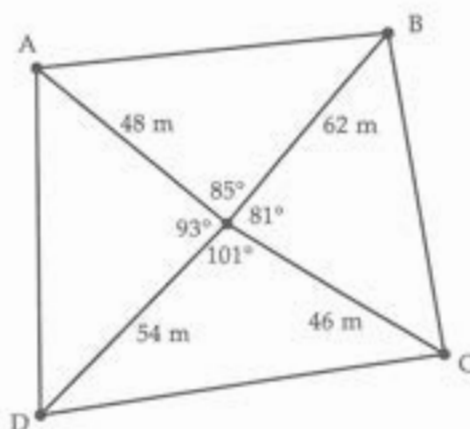
8 $a = 38$ cm, $b = 28$ cm and $c = 16$ cm

9 The two equal sides of an isosceles triangle are 165 mm long and the two equal angles are 42° . Find the length of the third side of the triangle, to the nearest millimetre.

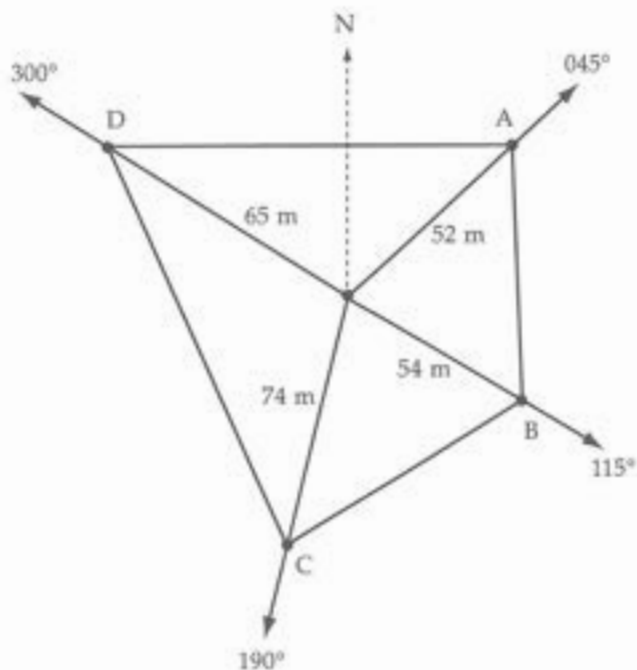
10 Find the size of the largest angle in a triangle with sides measuring 56 mm, 64 mm and 75 mm. (Hint: The largest angle is opposite the longest side.)

11 A balloon is flying in a straight line at an altitude of 600 metres. Matt is standing on the ground. He measures the angle of elevation of the balloon to be 85° . Three minutes later, Matt observed the balloon at an angle of elevation of 20° . How far has the balloon travelled in this time?

12 The piece of land represented in this diagram is based on a radial survey. Calculate its perimeter and area. Round your final answers to the nearest whole number.



13 A radial survey was conducted on a piece of land. A compass was used to determine the directions of the corner posts. The diagram shows all the measurements from the survey. Calculate the perimeter and the area of this piece of land. Round your final answers to the nearest whole number.



1 Review: trigonometry of right-angled triangles

A 1 17.5 cm 2 24.7 cm 3 10.9 cm 4 18.5 cm
5 49.3 cm 6 7.6 cm 7 32° 8 56° 9 67° 10 55°
11 66° 12 19°

B 1 17.4 cm 2 39.7 cm 3 26.1 cm 4 2.1 m 5 5.6 m
6 2.2 m 7 19°11' 8 60°17' 9 36°20' 10 36°52'
11 42°34' 12 56°50'

C 1 2.4 m 2 89.5 m 3 117 mm 4 9.2 m 5 5°
6 33.0 m 7 71°47' 8 112 m Riddle: Frostbite

2 Exact ratios for 30°, 45° and 60°

A 1 1 2 0.5 3 0.5 4 0.707 5 0.707 6 1.732 7 0.866
8 0.866 9 45° 10 45° 11 45° 12 60° 13 60° 14 30°
15 30° 16 45°

B 1 $\frac{\sqrt{3}}{2}$ 2 $\frac{1}{2}$ 3 $\sqrt{3}$ 4 1 5 $\frac{\sqrt{3}}{2}$ 6 $\frac{1}{2}$ 7 $\frac{1}{2}$ 8 $\frac{1}{2}$

C 1 30° 2 60° 3 45° 4 45° 5 60° 6 30° 7 30° 8 60°
Riddle: Square roots

3 Ratios for angles greater than 90°

A 1 0.466 2 0.602 3 0.966 4 0.900 5 0.961 6 2.145
7 0.883 8 0.829 9 2.747 10 0.375 11 0.675
12 0.695 13 0.602 14 0.276 15 8.144 16 0.799

B 1 0.819 2 -0.259 3 -1.732 4 -0.906 5 0.342
6 -1.192 7 -0.174 8 0.996 9 -0.669 10 0.906
11 -3.732 12 0.469 13 0.643 14 -0.588 15 -0.344
16 -0.174 17 -0.445 18 0.996 19 -0.829 20 0.956
21 0.682 22 -0.951 23 -1.664 24 -0.052 25 0.709
26 -4.247 27 -0.824 28 0.489 29 -0.240 30 0.989
31 -0.273 32 -0.333 33 0.792 34 -0.589 35 -0.511
36 0.977 37 -3.442 38 0.903 39 -0.994 40 -3.785

C 1 -0.342 2 -0.966 3 5.671 4 0.574 5 -0.707
6 -1.111 7 -0.866 8 -0.174 9 0.342 10 -0.574
11 1.192 12 -0.5 13 -0.966 14 4.011 15 0.966
16 0.466 17 0.208 18 -0.292 19 -0.407 20 0.777
21 -0.194 22 -0.559 23 -0.978 24 -1.483 25 0.906
26 -0.602 27 -0.900 28 0.276 29 -1.963 30 -0.940
31 0.602 32 -0.990 33 146° 34 134° 35 161° 147°
36 153° 37 104° 38 130° 39 112° 40 136° 41 147°
42 140° 43 129° 44 122° 45 113° 46 115° 47 151°
48 123° 49 112° 50 142° 51 101° 52 111° 53 143°
54 108° 55 118° 56 116° 57 155° 58 155° 59 156°
60 100° 61 125° 62 106° 63 168° 64 144°

4 The sine rule: finding the length of a side

A 1 0.83 2 0.98 3 0.75 4 0.90 5 0.97 6 0.78 7 0.89
8 0.64 9 6.57 10 3.01 11 27.93 12 3.64 13 83.69
14 4.70 15 41.06 16 3.62

B 1 31.9 cm 2 32.4 cm 3 19.7 cm 4 51.2 cm
5 28.1 cm 6 47.5 cm 7 32.1 cm 8 32.6 cm
9 46.8 cm 10 40.6 cm 11 20.9 cm 12 94.1 cm

C 1 58.3 cm 2 34.8 cm 3 49.1 cm 4 76.2 cm
5 37.5 cm 6 39.5 cm 7 3.9 m 8 14.9 cm 9 5.5 m
10 7.6 m 11 7.2 m 12 6.7 m 13 22.8 m 14 9.6 m
15 14.1 m

5 The sine rule: finding the size of an angle

A 1 66° 2 26° 3 49° 4 13° 5 35° 6 11° 7 44° 8 76°
9 125° 10 95° 11 140° 12 155° 13 108° 14 164°
15 161° 16 128°

B 1 75° 2 47° 3 61° 4 55° 5 70° 6 55° 7 120°
8 112° 9 110° 10 141° 11 96° 12 132°

C 1 48°55' 2 46°53' 3 51°37' 4 14°59' 5 53°11'
6 100°17' 7 37°56' 8 37°51' 9 63°36' 10 33°34'
11 92°56' 12 57°37'

6 The sine rule: mixed questions

A 1 25.7 cm 2 18.8 cm 3 20.4 cm 4 18.8 cm
5 30.8 cm 6 26.6 cm 7 68° 8 60° 9 75° 10 69°
11 133° 12 75°

B 1 1.6 m 2 2.8 m 3 3.3 m 4 2.9 m 5 2.5 m 6 3.6 m
7 47°19' 8 25°26' 9 75°42' 10 61°19' 11 134°46'
12 101°51'

C 1 11.7 cm 2 18.8 cm 3 23.4 cm 4 30.7 cm 5 41°26'
6 69°6' 7 41°47' 8 57°20' 9 426 m 10 19°

7 The cosine rule: finding the length of a side

A 1 0.14 2 0.89 3 0.96 4 0.92 5 -0.31 6 -0.86
7 -0.53 8 -0.95 9 13.52 10 -4.08 11 27.89
12 -19.74 13 78.16 14 -14.95 15 -60.53 16 72.89

B 1 14.7 cm 2 29.7 cm 3 3.4 cm 4 69.5 cm
5 14.0 cm 6 9.4 cm 7 56.1 cm 8 50.8 cm 9 8.6 cm
10 21.5 cm 11 25.5 cm 12 54.4 cm

C 1 7.5 cm 2 48.6 cm 3 12.2 cm 4 36.5 cm 5 8.4 m
6 4.8 m 7 16.7 m 8 30.2 m 9 145.2 m 10 8.8 m
11 24.6 m 12 3.3 m 13 2.4 cm 14 138.9 cm
15 32.3 cm

8 The cosine rule: finding the size of an angle

A 1 50° 2 61° 3 83° 4 75° 5 68° 6 85° 7 24° 8 36°
9 156° 10 132° 11 139° 12 174° 13 122° 14 92°
15 113° 16 121°

B 1 76° 2 79° 3 23° 4 85° 5 22° 6 87° 7 111°
8 128° 9 130° 10 121° 11 115° 12 104°

- C** 1 $38^{\circ}43'$ 2 $27^{\circ}42'$ 3 $84^{\circ}39'$ 4 $132^{\circ}30'$ 5 $52^{\circ}32'$
 6 $20^{\circ}27'$ 7 $44^{\circ}25'$ 8 $77^{\circ}20'$ 9 $96^{\circ}15'$ 10 $31^{\circ}21'$
 11 $146^{\circ}33'$ 12 $88^{\circ}36'$ 13 $102^{\circ}50'$ 14 $112^{\circ}53'$
 15 $73^{\circ}42'$

9 The cosine rule: mixed questions

- A** 1 11.5 cm 2 10.4 cm 3 14.0 cm 4 43.2 cm
 5 22.8 m 6 6.6 m 7 62° 8 86° 9 41° 10 39°
 11 53° 12 64°
- B** 1 17.6 cm 2 11.7 cm 3 3.7 m 4 11.4 m 5 77.2 mm
 6 9.1 m 7 $111^{\circ}38'$ 8 $23^{\circ}47'$ 9 $84^{\circ}55'$ 10 $32^{\circ}31'$
 11 $87^{\circ}15'$ 12 $48^{\circ}51'$
- C** 1 9.1 cm 2 19.5 cm 3 9.9 m 4 39.7 m 5 $55^{\circ}4'$
 6 $126^{\circ}25'$ 7 $109^{\circ}29'$ 8 $69^{\circ}31'$

10 Mixed questions: the sine and cosine rules

- A** 1 7.4 cm 2 29.1 cm 3 28.1 cm 4 10.5 cm
 5 43.0 cm 6 185.0 cm 7 15.1 m 8 54.1 cm
 9 201.5 mm
- B** 1 39° 2 57° 3 31° 4 40° 5 48° 6 63° 7 45° 8 25°
 9 12°
- C** 1 8.1 m 2 $57^{\circ}30'$ 3 $18^{\circ}43'$ 4 20.4 cm 5 $67^{\circ}57'$
 6 $72^{\circ}58'$ 7 $82^{\circ}54'$ 8 29.5 m 9 8.5 m 10 $60^{\circ}0'$
 11 $105^{\circ}49'$ 12 7.5 m

11 Problems involving two triangles

- A** 1 10.5 cm 2 31.6 cm 3 25.8 cm 4 71° 5 54° 6 33°
- B** 1 18.3 m 2 BD = 129.1 m, CD = 107.1 m 3 81 m
 4 AB = 60.7 m, BD = 36.5 m 5 44.7 m 6 155 m
 7 BD = 37.1 m, BC = 48.9 cm 8 BD = 18.6 cm,
 A = 91°
- C** 1 267.1 m, 185.5 m 2 3556 m 3 132 cm, 146 cm
 4 274.2 m 5 35.5 cm 6 63°

12 Review: the sine rule and the cosine rule

- A** 1 20.8 cm 2 23.9 cm 3 27.1 cm 4 45° 5 60°
 6 125° 7 18.6 cm 8 22.0 cm 9 13.1 cm 10 47°
 11 48° 12 97°
- B** 1 3.6 m 2 5.6 m 3 2.6 m 4 $52^{\circ}43'$ 5 $112^{\circ}2'$
 6 $50^{\circ}11'$ 7 141 mm 8 124 mm
- C** 1 9.1 cm 2 22.6 cm 3 27.0 cm 4 13.0 cm
 5 $35^{\circ}2'$ 6 $63^{\circ}36'$ 7 $58^{\circ}46'$ 8 $119^{\circ}30'$ 9 47°
 10 32 km 11 5.2 m 12 $51^{\circ}12'$

13 The area of a triangle

- A** 1 19° 2 75° 3 60° 4 75° 5 88° 6 40° 7 53° 8 42°
 9 1.563 10 7.502 11 9.798 12 71.629
- B** 1 194.9 cm² 2 145.9 cm² 3 37.9 cm² 4 113.8 cm²
 5 1107.6 cm² 6 474.1 cm² 7 13.4 cm² 8 100.4 cm²
 9 42.0 m² 10 7.2 m² 11 138.5 m² 12 222.2 m²
- C** 1 68.52 cm² 2 92.25 cm² 3 40.14 m² 4 5.53 m²
 5 0.87 m² 6 8.05 m² 7 4115.71 m² 8 95.41 m²
 9 27.80 m² 10 114.37 m² 11 63.26 m² 12 14.61 m²
 13 291.84 m² 14 10.57 m² 15 6.26 m²

14 Practical application: radial surveys

- A** 1 1.7 m 2 1.7 m 3 2.7 m 4 73° 5 47° 6 53°
 7 4.3 m² 8 8.0 m² 9 3.2 m²
- B** 1 a 85.4 m b BC = 91.9 m, CD = 36.9 m, DA = 74.2 m
 c 288 m d 1495.9 m² e BPC = 1041.1 m², CPD = 755.0 m², APD = 1322.2 m² f 4615 m² 2 perimeter = 385 m, area = 8975 m² 3 perimeter = 348 m, area = 7142 m²
- C** 1 a 115° b 87° c 101.3 m d 331 m e 6669 m²
 2 perimeter = 385 m, area = 9003 m² 3 perimeter = 347 m, area = 7496 m²

15 Mixed problems

- A** 1 19.8 cm 2 41.4 cm 3 25.4 cm 4 43° 5 116°
 6 108° 7 323.1 cm² 8 332.3 cm² 9 562.7 cm²
- B** 1 42 m² 2 15° 3 20.5 km 4 2.1 m 5 53 cm 6 122°
 7 3402 cm² 8 19.1 km
- C** 1 91 mm 2 166 m 3 791.5 m 4 23.4 cm² 5 28.6 km
 6 $60^{\circ}21'$ 7 8.2 km 8 172 873 mm² 9 26.4 n. miles
 10 207°

16 The graph of $y = \sin x^{\circ}$

- A** 1 0, 0.5, 0.87, 1, 0.87, 0.5, 0, -0.5, -0.87, -1, -0.87,
 -0.5, 0
 0, -0.5, -0.87, -1, -0.87-0.5, 0, 0.5, 0.87, 1, 0.87, 0.5, 0
- B** 1 1 2 0 3 -1 4 270° 5 30° , 150° 6 90° 7 60° ,
 120° 8 0, 180° , 360° 9 210° , 330°
- C** 4 a $y = -2 \sin x$ b $y = \sin 2x$ c $y = 3 \sin x$
 d $y = \sin \frac{1}{2}x$ 5 3 6 360° 7 360° 8 120° 9 -1

17 The graph of $y = \cos x^{\circ}$

- A** 1 1, 0.87, 0.5, 0, -0.5, -0.87, -1, -0.87, -0.5, 0, 0.5,
 0.87, 1
 1, 0.87, 0.5, 0, -0.5, -0.87, -1, -0.87, -0.5, 0, 0.5, 0.87, 1

B 1 1 2 0 3 0 4 0° , 360° 5 60° , 300° 6 180° 7 0.5
8 -0.5 9 0.5

C 1 1, 0.87, 0.71, 0.5, 0, -0.5, -0.71, -1, -0.71, 0, 1
2 2, 1.73, 1.41, 1, 0, -1, -1.41, -2, -1.41, 0, 2
3 1, 0.5, 0, -0.5, -1, -0.5, 0, 1, 0, -1, 1
4 -1, -0.87, -0.71, -0.5, 0, 0.5, 0.71, 1, 0.71, 0, -1
5 a $y = 2 \cos x$ b $y = \cos 2x$ c $y = \cos \frac{1}{2}x$
d $y = -\cos x$ 6 1 7 360° 8 3 9 180°

18 The graph of $y = \tan x^\circ$

A 1 0, 0.58, 1, 1.73, undefined, -1.73, -1, -0.58, 0, 0.58,
1, 1.73, undefined, -1.73, -1, -0.58, 0
B 1 continuous 2 -270° , -90° , 90° , 270° 3 180° , 180°
4 amplitude 5 asymptotes 6 -360° , -180° , 0° , 180° ,
 360° 7 -315° , -135° , 45° , 225° 8 -225° , -45° , 135° ,
 315°
C 1 180° 2 $x = 90^\circ$, $x = 270^\circ$ 3 60° 4 $x = -30^\circ$,
 $x = 30^\circ$, $x = 90^\circ$ 5 0, 3, -3, 0, 3, -3, 0 6 0, 1,
undefined, -1, 0, 1, undefined, -1, 0 7 $y = \tan 2x$

19 Further trigonometric graphs

A 1 a = 0.866, b = 90° , c = 0.5, d = -0.5, e = 330° , f = 0
3 a $x = 45^\circ$, 135° b $x = 225^\circ$, 315° c $x = 210^\circ$, 330°
d $x = 0^\circ$, 180° , 360°
B 1 a 90° , 270° b infinite c 107° , 287° d 215°
2 A(90° , 3), B(180° , 0), C(270° , -3), D(360° , 0) 3 p = 3,
q = 1, m = 2, n = $\frac{1}{2}$ ($y = 3 \sin x$ and $y = 2 \cos \frac{1}{2}x$);
 $y = 3 \sin x$ has period 360° , amplitude = 3;
 $y = 2 \cos \frac{1}{2}x$ has period 720° , amplitude = 2
C 2 $\cos x$: 1, 0.5, 0, -0.5, -1, -0.5, 0, 0.5, 1
 $\sec x$: 1, 2, undefined, -2, -1, -2, undefined a 360°
b infinite c infinite
3 $\sin x$: 0, 0.5, 1, 0.5, 0, -0.5, -1, -0.5, 0
 $\operatorname{cosec} x$: undefined, 2, 1, 2, undefined, -2, -1, -2,
undefined a 360° b infinite c infinite

20 Simple trigonometric equations

A 1 37° 2 67° 3 22° 4 25° 5 73° 6 41° 7 57°
8 30° 9 39° 10 58° 11 67° 12 44° 13 48° 14 51°
15 57° 16 89° 17 73° 18 84° 19 69° 20 6° 21 67°
22 62° 23 32° 24 74° 25 28° 26 25° 27 19° 28 29°
B 1 143° 2 166° 3 121° 4 175° 5 102° 6 115°
7 139° 8 99° 9 153° 10 129° 11 117° 12 108°
13 30° , 150° 14 44° , 136° 15 59° , 121° 16 13° , 167°
17 65° 18 120° 19 136° 20 30° 21 27° 22 124°
23 68° 24 126° 25 24° , 156° 26 60° 27 45° 28 53° ,
 127° 29 107° 30 117° 31 30° , 150° 32 72° 33 48°

34 112° 35 120° 36 49° , 131°

C 1 60° , 300° 2 210° , 330° 3 56° , 236° 4 135° , 315°
5 117° , 297° 6 229° , 311° 7 58° , 238° 8 201° , 338°
9 38° , 322° 10 120° , 300° 11 108° , 208° 12 233° ,
 307° 13 90° 14 63° , 243° 15 180° 16 41° , 319°
17 270° 18 50° , 230° 19 42° , 138° 20 27° , 207°
21 71° , 289° 22 127° , 307° 23 30° , 150° 24 124° ,
 304°

21 Trigonometric relationships

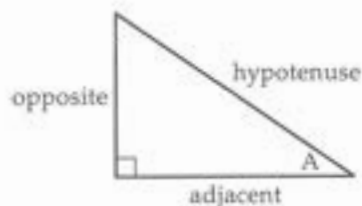
A 1 true 2 true 3 true 4 true 5 true 6 true 7 true
8 true

22 Review: trigonometry

A 1 9.9 cm 2 6.6 cm 3 33.9 cm 4 $44^\circ 54'$ 5 $34^\circ 23'$
6 $54^\circ 19'$ 7 31.9 cm 8 31.3 cm 9 31.1 cm 10 82°
11 61° 12 40° 13 136 cm^2 14 321 cm^2 15 234 cm^2
B 1 100° 2 103° 3 118° 4 3.8 m 5 4.7 m 6 1.9 m
7 $91^\circ 28'$ 8 $58^\circ 16'$ 9 $93^\circ 12'$ 10 902 cm^2 11 401 m
12 217 mm 13 347.4 m 14 6924 mm
15 AB = 56.0 m, BD = 37.5 m
C 1 23.9 cm 2 27.6 cm 3 19.3 cm 4 40.5 cm 5 $45^\circ 11'$
6 $54^\circ 31'$ 7 $35^\circ 5'$ 8 $116^\circ 48'$ 9 245 mm 10 $77^\circ 3'$
11 186.7 m 12 perimeter = 298 m, area = 5404 m^2
13 perimeter = 348 m, area = 7142 m^2

Formulae

Right-angled triangles

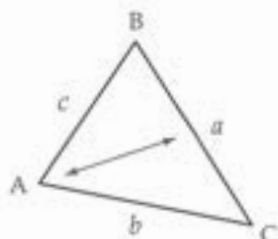


$$\sin A = \frac{\text{opposite side}}{\text{hypotenuse}}$$

$$\cos A = \frac{\text{adjacent side}}{\text{hypotenuse}}$$

$$\tan A = \frac{\text{opposite side}}{\text{adjacent side}}$$

Other triangles



The Sine Rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

or $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$

The Cosine Rule

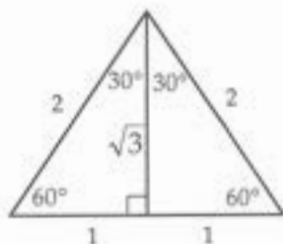
$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

The Area Formula

$$\text{Area} = \frac{1}{2}ab \sin C$$

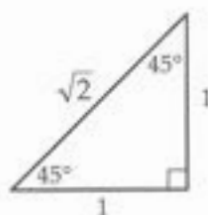
Exact ratios of special angles



$$\sin 45^\circ = \frac{1}{\sqrt{2}}$$

$$\cos 45^\circ = \frac{1}{\sqrt{2}}$$

$$\tan 45^\circ = 1$$



$$\sin 30^\circ = \frac{1}{2}$$

$$\cos 30^\circ = \frac{\sqrt{3}}{2}$$

$$\tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$\cos 60^\circ = \frac{1}{2}$$

$$\tan 60^\circ = \sqrt{3}$$

Instant Lessons

Quality ready-to-use resources

Each book in the **Instant Lessons** series is a collection of fresh and versatile worksheet activities which may be photocopied for student use. They are unique in providing **the highest quality learning outcomes with instant usability.**

This checklist highlights the major uses and benefits of these books.

Versatile uses		Teaching benefits	
Supplementary classroom lessons	✓	Creative teaching ideas	✓
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Extension work for advanced students	✓	A way of targeting student weaknesses	✓
Preparation for assessment	✓	Suitable for individual or group work	✓
Exam revision	✓	A variety of activities for home study	✓

Use them to supplement your existing resources, to extend student capabilities, for revision purposes, or for class work when the regular teacher is absent.

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Topics include:

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Exact ratios for 30° , 45° and 60°
Ratios for angles greater than 90°
The sine rule: finding the length of a side and the size of an angle
The cosine rule: finding the length of a side and the size of an angle

Problems involving two triangles
Area of a triangle
Practical application of trigonometry:
radial surveys
Graphs of $y = \sin x$, $y = \cos x$ and $y = \tan x$
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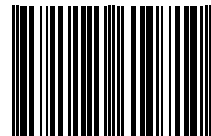
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ISBN: 978-1-92561-071-0



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